

03.24.2026 **Wayne Lin, FAE**



/ Agenda

Introductions

Get To Know MCXN947

- GUI Guider Software download
- GUI Guider Install

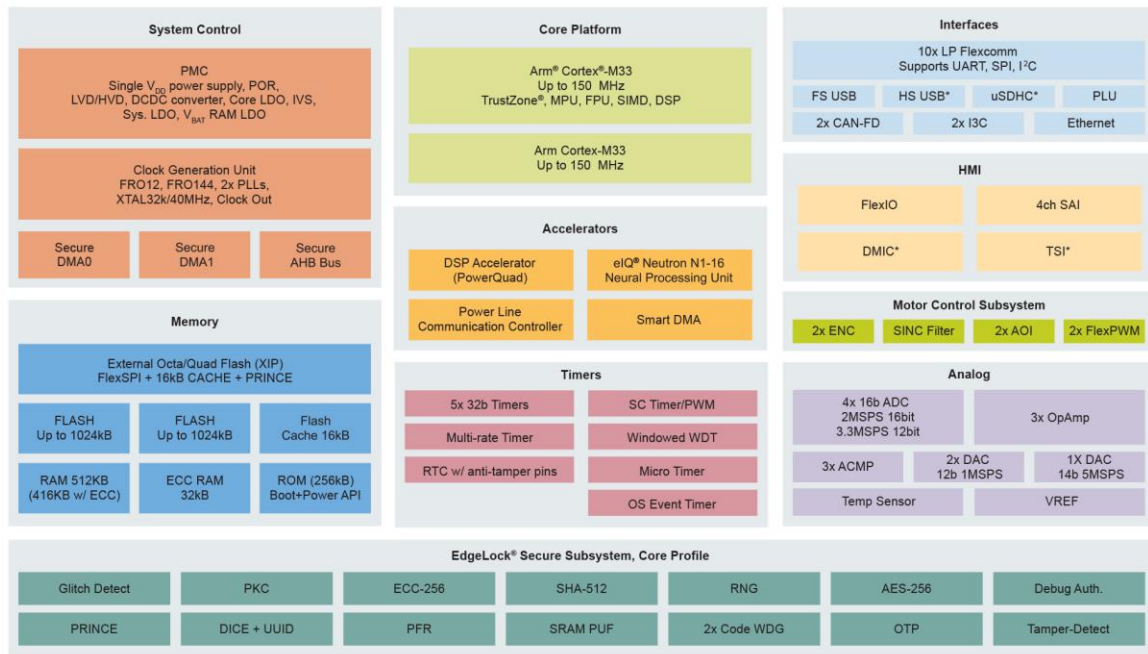
Install Tool

- MCUXPresso / GUI Guider / TeraTerm

GUI Guider Hands On

Get To Know MCXN947

Key Features and Benefits



Key Feature Highlights

Performance

- Dual CM33 150MHz
- DSP, NPU

Intelligent Peripherals

- Flexcomm, FlexIO, FlexPWM, FlexSPI
- Touch, Motor Control, Analog

Advanced Security

- Crypto, Tamper-Detect
- Life-cycle management

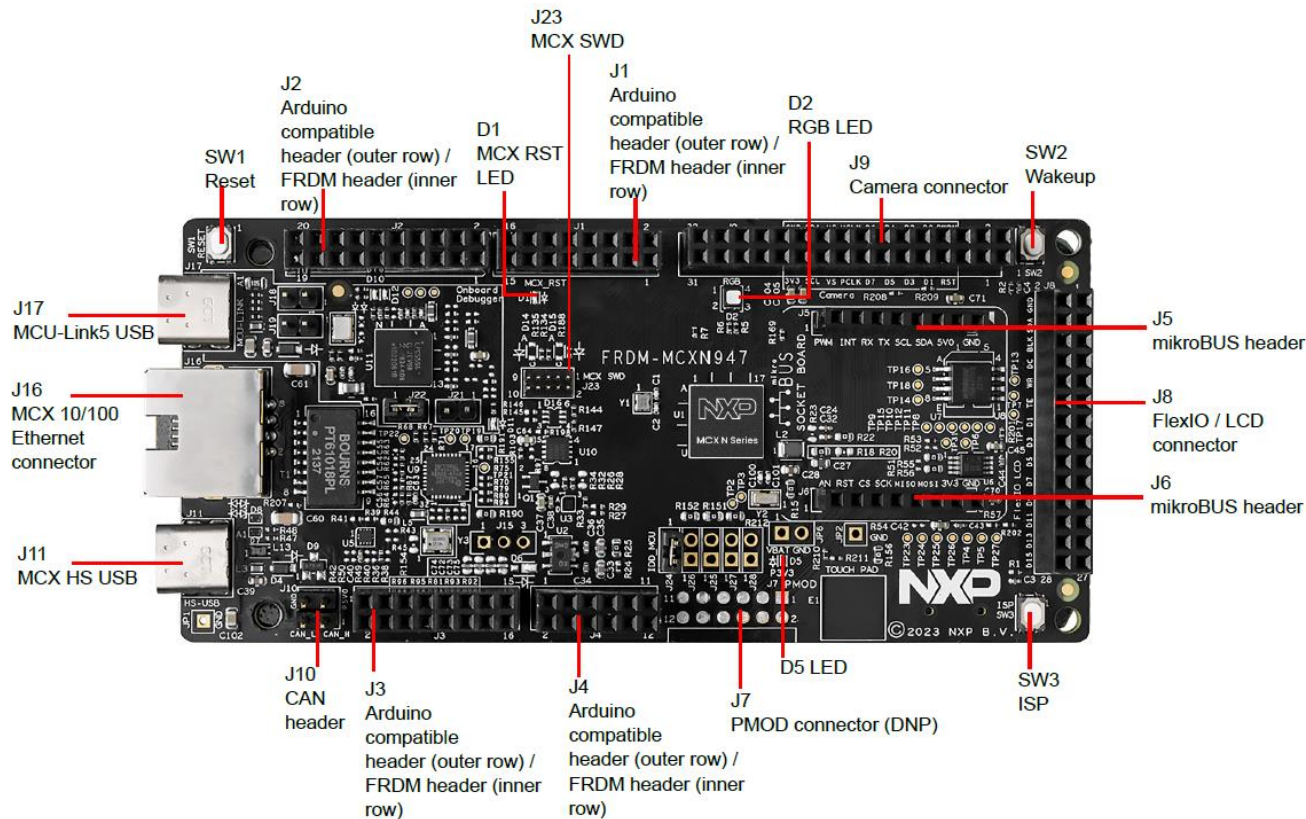
Power and Operation

- Buck DC-DC, Core LDO, other LDOs
- Separate AO domain on VBAT pin
- Operating voltage: 1.71V to 3.6V
- Temperature range: -40 to 105 °C
- IOs: 1.71V to 3.6V

Package Options

- VFBGA184: 9 x 9 x 0.86 mm, 0.5 mm pitch
- HLQFP100: 14 x 14 x 1.4 mm, 0.5 mm pitch

FRDM-MCXXN947



Install MCUXpresso

/ Download MCUXpresso IDE

Download MCUXpresso IDE

DEVELOPMENT IDES AND BUILD TOOLS

**MCUXpresso Integrated Development Environment
(IDE)** UPDATED

FLEXERA Rev All Jan 14, 2025 1 KB MCUXPRESSO

DOWNLOAD



/ Download MCUXpresso 24.12

DEVELOPMENT IDES AND BUILD TOOLS

MCUXpresso Integrated Development Environment (IDE) UPDATED

FLEXERA Rev All Jan 14, 2025 1 KB MCUXPRESSO

1

DOWNLOAD

Current

Previous

2

Version	Description	Date Available	
24.12	MCUXpresso IDE	Jan 10, 2025	Download Log

/ Download MCUXpresso 24.12

Software Terms and Conditions

MCUXpresso IDE

Please read the following agreement and click "I AGREE" at the bottom before downloading your software.

LA_OPT_NXP_Software_License v59 December 2024

IMPORTANT. Read the following NXP Software License Agreement ("Agreement") completely. By selecting the "I Accept" button at the end of this page, or by downloading, installing, or using the Licensed Software, you indicate that you accept the terms of the Agreement, and you acknowledge that you have the authority, for yourself or on behalf of your company, to bind your company to these terms. You may then download or install the file. In the event of a conflict between the terms of this Agreement and any license terms and conditions for NXP's proprietary software embedded anywhere in the Licensed Software file, the terms of this Agreement shall control. If a separate license agreement for the Licensed Software has been signed by you and NXP, then that agreement shall govern your use of the Licensed Software and shall supersede this Agreement.

[NXP SOFTWARE LICENSE AGREEMENT](#)

This is a legal agreement between your employer, of which you are an authorized representative, or, if you have no

I Agree

Cancel

/ Download MCUXpresso 24.12

依照使用系統下載相應版本

MCUXpresso IDE

Files

License Keys

Notes

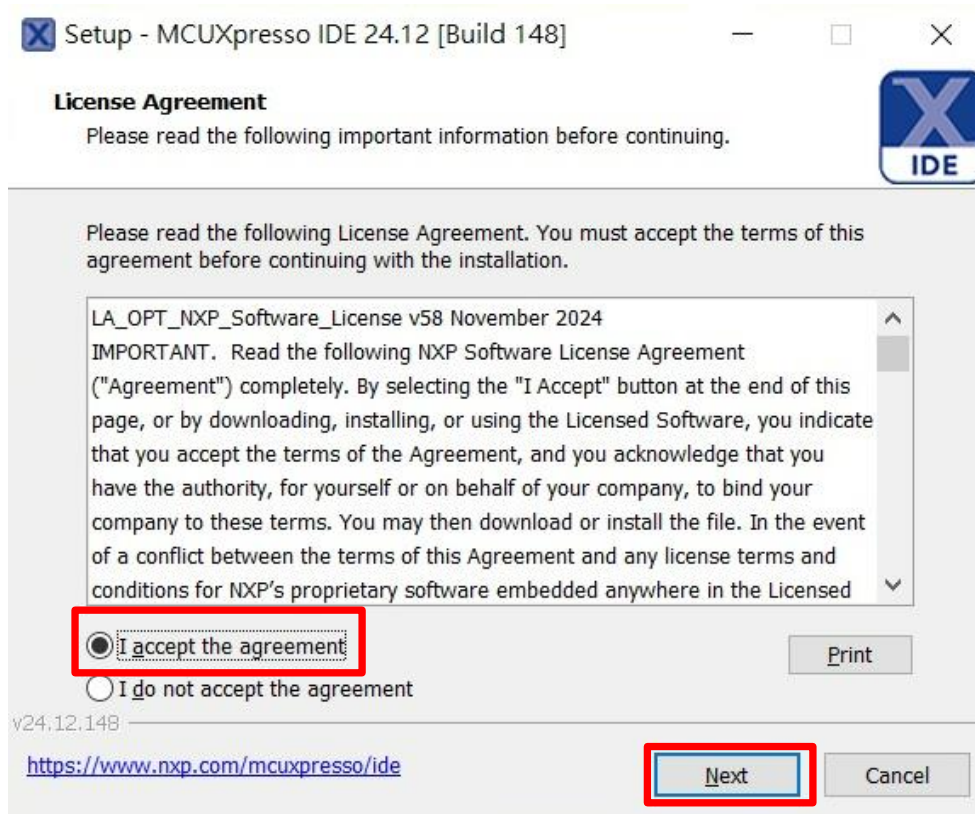
[Download Help](#)

Show All Files

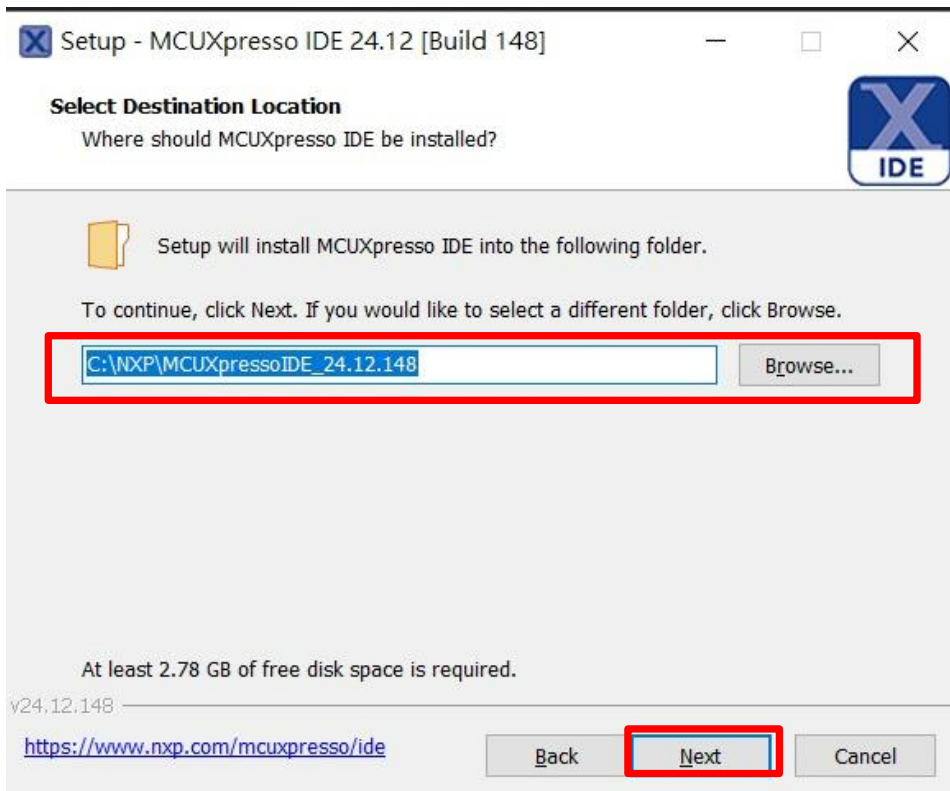
4 Files

+	File Description	File Size	File Name
+	MCUXPRESSO_V24.12 - Linux	1.4 GB	mcuxpressoide-24.12.148.x86_64.deb.bin
+	MCUXPRESSO_V24.12 - MacOSX - 86-64	1.1 GB	MCUXpressoIDE_24.12.148.x86-64.pkg
+	MCUXPRESSO_V24.12 - MacOSX - aarch64	1.1 GB	MCUXpressoIDE_24.12.148.aarch64.pkg
+	MCUXPRESSO_V24.12 - Windows	1.2 GB	MCUXpressoIDE_24.12.148.exe

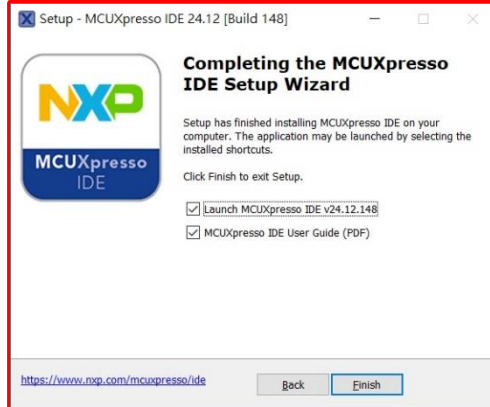
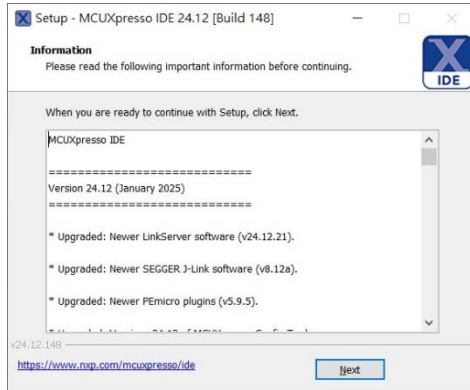
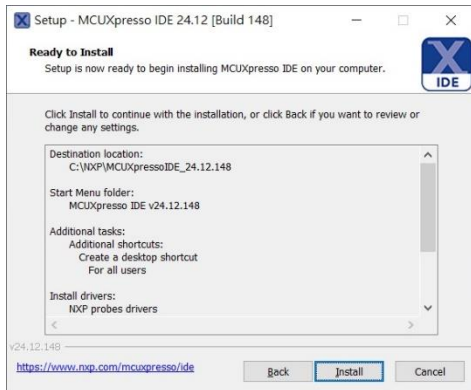
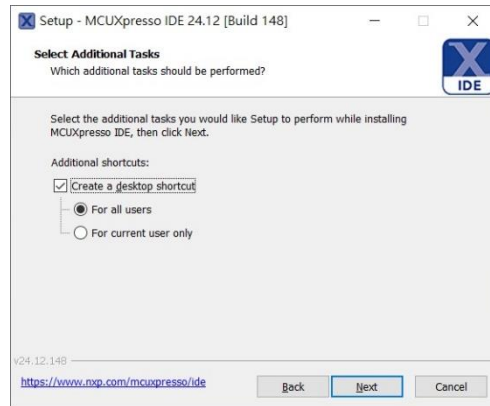
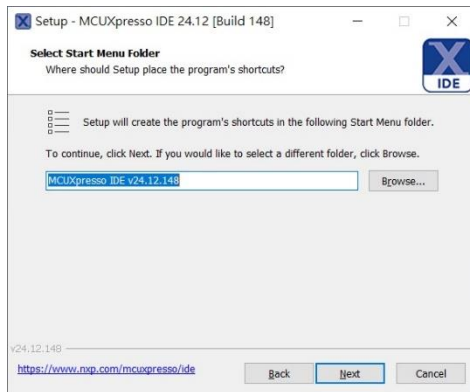
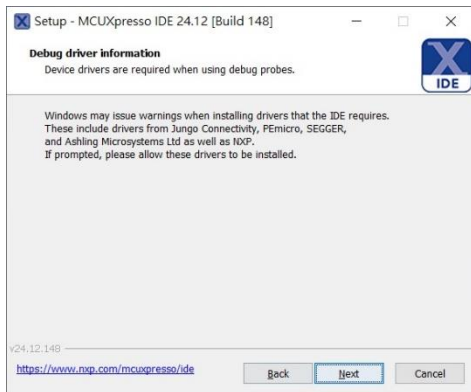
/ Install MCUXpresso IDE



/ Install MCUXpresso IDE



Install MCUXpresso IDE



/ Install MCUXpresso IDE



目標：加入 **-showlocation -nl en_US**
可以將IDE改成英文版本

MCUXpresso IDE
v11.3.1_5262

MCUXpresso SDK

/ Download MCUXpresso SDK



LANGUAGE

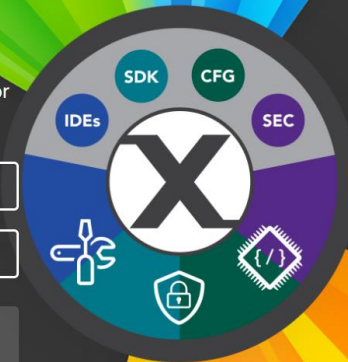
SUPPORT

MCUXpresso SDK Builder

The MCUXpresso SDK brings open source drivers, middleware, and reference example applications to speed your software development. Customize and download an SDK specific to your processor or evaluation board selections.

 Select Development Board Access My SDK Dashboard Expansion Board Hub Application Code Hub

 NXP strongly recommends you update to the latest version of MCUXpresso SDK that contains essential security updates for Bluetooth Low Energy software stack implementations.



<https://mcuxpresso.nxp.com/en/welcome>

/ Download MCUXpresso SDK

MCUXpresso SDK Builder

SDK Dashboard

Select Board / Processor

Filters

Middleware

Examples

Toolchains

Processor Parametrics

EXPLORE

Expansion Board Hub

Application Code Hub

DOWNLOADS

MCUXpresso IDE

MCUXpresso for VS Code

MCUXpresso Secure Provisioning Tool

MCUXpresso Config Tools

Offline data

Hardware

Off

Middleware

0

Examples

0

Toolchains

Off

Parametrics

0

CLEAR

mi

> LPC

> MCX

FRDM-MCXA153 (MCXA153)

FRDM-MCXA156 (MCXA156)

FRDM-MCXC041 (MCXC041)

FRDM-MCXC242 (MCXC242)

FRDM-MCXC444 (MCXC444)

FRDM-MCXN236 (MCXN236)

FRDM-MCXN947 (MCXN947)

FRDM-MCXW71 (MCXW716CxxxA)

MCX-N5XX-EVK (MCXN547)

MCX-N9XX-EVK (MCXN947)

> MW

> QN

Select

FRDM-MCXN947

FRDM-MCXN947

App

Board

CMSIS



Found

793 HW solutions

157 Boards

106 Kits

530 Processors

+ ADD HW TO FILTERS

Explore selection with

PINS TOOL

CLOCKS TOOL

24.12.00

BUILD SDK

選擇 FRDM-MCX947(MCXN947)

/ Download MCUXpresso SDK

Developer Environment Settings

Selections here (operating host system SDK and Generated Projects)

Host OS: Windows

Toolchain / IDE: MCUXpresso IDE Search...

☒	Name	Description	Dependencies
☒	AWS IoT Core	FreeRTOS Amazon Web Service (AWS) IoT Core SDK	 
☒	CMSIS DSP Library	CMSIS DSP Software Library	
☒	coreHTTP	coreHTTP	 
☒	elQ	 elQ machine learning (ML) middleware includes the ...	 
☒	emWin	emWin graphics library	
☒	Ethermind BT/BLE Stack	FreeRTOS nxp_bt_ble_stack	 
☒	Fatfs	FAT File System stack	
☒	FreeMASTER	FreeMASTER communication driver for 32bit platforms	
☒	FreeRTOS	FreeRTOS Real-time operating system for microco...	

Download SDK

Build SDK for
FRDM-MCXN947

FRDM-MCXN947



SDK Version: 24.12.00 (released 2025-01-14)

Additional Information

Board	FRDM-MCXN947
Processor	MCXN947
Core Type / Max Freq	Cortex-M33 / 150 MHz
	Cortex-M33 / 150 MHz
Device Memory Size	2048 KB Flash / 512 KB RAM

Tested toolchain version

BUILD SDK

/ Download MCUXpresso SDK

MCUXpresso SDK Dashboard

Access, Download, and Share your SDK Builds.



SDK_24.12.00_FRDM-MCXX947

2025-02-17

Remove

Windows

24.12.00

FRDM-MCXX947

Rebuild

MCUXpresso IDE

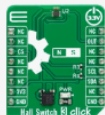
Config Tools

CMSIS DSP Library, FreeRTOS, TinyCBOR, PKCS #11, multicore, Ilhttp, FreeMASTER, USB Type-C PD Stack, USB Host, Device, OTG Stack, LVGL, LittleFS, Fatfs, emWin, coreHTTP, NXP Wi-Fi, TF-M, PSA Test Suite, Mbed Crypto, NXP IoT Agent, MCUBoot, mbedTLS 3.X, mbedTLS, VoiceSeeker (no AEC), lwIP, AWS IoT Core, NXP PSA CRYPTO DRIVER, NXP ELS PKC, SDMMC Stack, Motor Control Software, Ethermind BT/BLE Stack, Secure Element Host Library, NXP Touch Library, eIQ

Share

[Add Description](#)

Download



/ Download MCUXpresso SDK

Downloads

MCUXpresso SDK

[Download SDK Archive \(114 MB\)](#)

MD5: 657d8c6d3069ea8b2c84f5ac43491279 [📋](#)

[Download Standalone Example Project](#)

[i](#) For MCUXpresso IDE, example projects can be imported as standalone projects directly within the IDE by downloading the SDK Archive

Additional Tools

[i](#) Download additional tools from NXP or its partners to create new projects and modify example projects using the associated software components included in this SDK

[🔗 MCUXpresso SDK for Motor Control](#)

Online Documentation

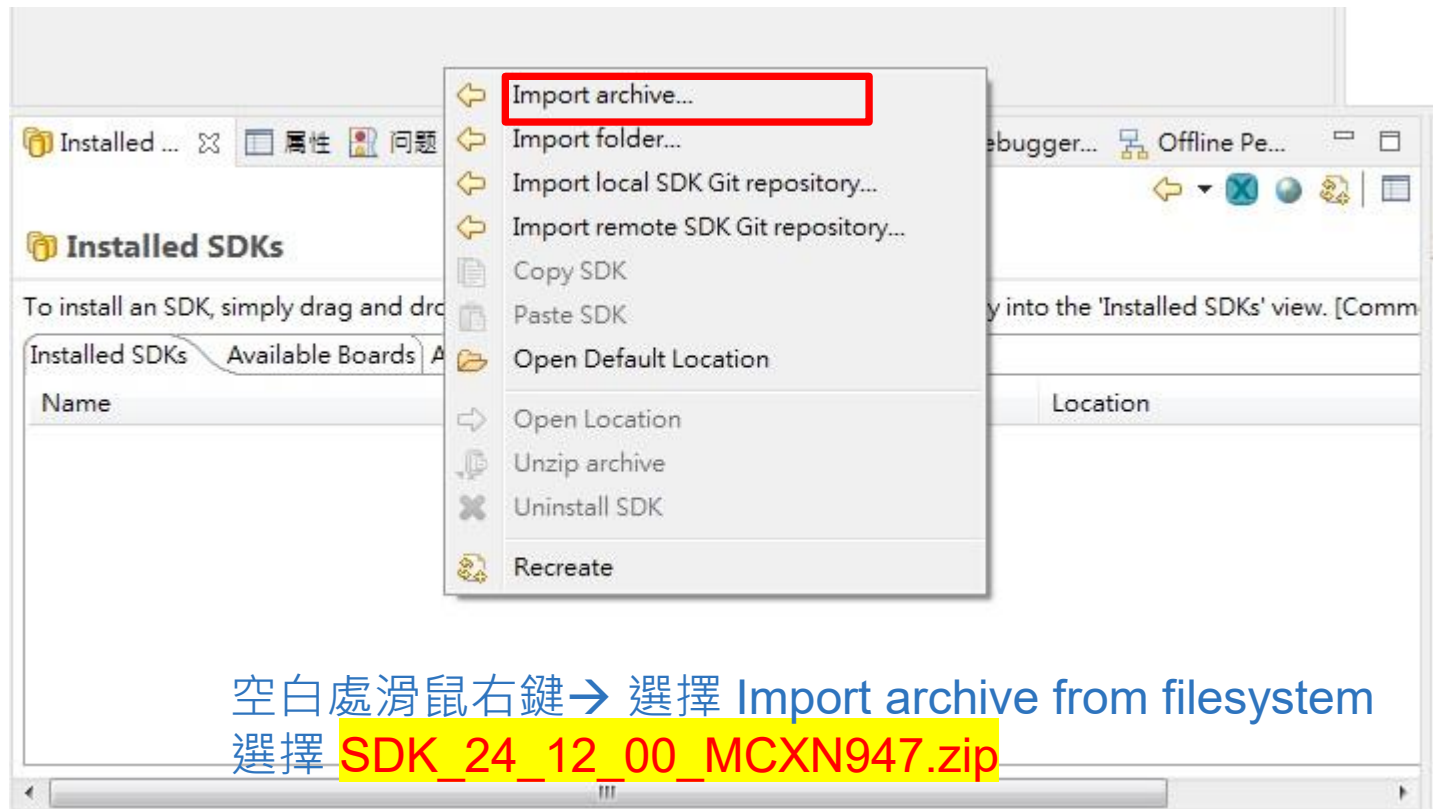
[📄 MCUXpresso SDK Documentation](#)

MCUXpresso Config Tools

[Download Config Tools](#)

[CLOSE](#)

/ Import SDK



Install GuiGuider

Download GUIGUIDER IDE

HMI DESIGN SOFTWARE

GUI Guider v1.8.1 Installer for Windows FEATURED

EXE Rev 1.8.1 Sep 30, 2024 430464 KB

GUI-GUIDER-SETUP-1.8.1-GA-WIND

 **Account Required**



/ Download GUI Guide

Agreement : GUI Guider v1.8.1 Installer for Windows

LA_OPT_NXP_Software_License v59 December 2024

IMPORTANT. Read the following NXP Software License Agreement ("Agreement") completely. By selecting the "I Accept" button at the end of this page, or by downloading, installing, or using the Licensed Software, you indicate that you accept the terms of the Agreement, and you acknowledge that you have the authority, for yourself or on behalf of your company, to bind your company to these terms. You may then download or install the file. In the event of a conflict between the terms of this Agreement and any license terms and conditions for NXP's proprietary software embedded anywhere in the Licensed Software file, the terms of this Agreement shall control. If a separate license agreement for the Licensed Software has been signed by you and NXP, then that agreement shall govern your use of the Licensed Software and shall supersede this Agreement.

NXP SOFTWARE LICENSE AGREEMENT

This is a legal agreement between your employer, of which you are an authorized representative, or, if you have no employer, you as an individual ("you" or "Licensee"), and NXP USA, Inc., if Licensee is located within the United States or NXP Semiconductors Netherlands B.V., if Licensee is located outside of the United States ("NXP"). It concerns your rights to use the software provided to you in binary or source code form and any accompanying written materials (the "Licensed Software"). The Licensed Software may include any updates or error corrections or documentation relating to the Licensed Software provided to you by NXP under this Agreement. In consideration for NXP allowing you to access the Licensed Software, you are agreeing to be bound by the terms of this Agreement. If you do not agree to all of the terms of this Agreement, do not download or install the Licensed Software. If you change your mind later, stop using the Licensed Software and delete all copies of the Licensed Software in your possession or control. Any copies of the Licensed Software that you have already distributed, where permitted, and do not destroy will continue to be governed by this Agreement. Your prior use will also continue to be governed by this Agreement.

I ACCEPT

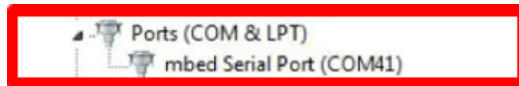
DECLINE

Install Tera Term

1. **Download** Tera Term from SourceForge. After the download, run the installer and then return to this webpage to continue.

[DOWNLOAD](#)

2. **Launch** Tera Term. The first time it launches, it will show you the following dialog. Select the serial option. Assuming your board is plugged in, there should be a COM port automatically populated in the list.



Not sure how to use a terminal application? Try one of these tutorials:

[Tera Term Tutorial](#)

[PuTTY Tutorial](#)

3. **Configure** the serial port settings (using the COM port number identified earlier) to 115200 baud rate, 8 data bits, no parity and 1 stop bit. To do this, go to Setup -> Serial Port and change the settings.

• **Download [Tera Term](#).** After the download, run the installer and then return to this webpage to continue.

Google Driver

2025/3/8

☰ Uart API.7z 👤



📄 01.MakeNTU 20260324_GuiGuider_V1.pdf

Download

Getting Started with GUI Guider

Step 0. 電腦先連上Internet



1. 點擊桌面GUI
Guider 開啟

2. 網路需要連線後登入NXP 帳號

NXP Authentication

NXP

Access your full NXP.com experience and stay up to date on our products and services.

Email Address*

wayne.lin@avnet.com

Password* [Show](#)

.....

[Forgot your password? Reset it.](#)

☒ Keep me signed in (uncheck if using public device).

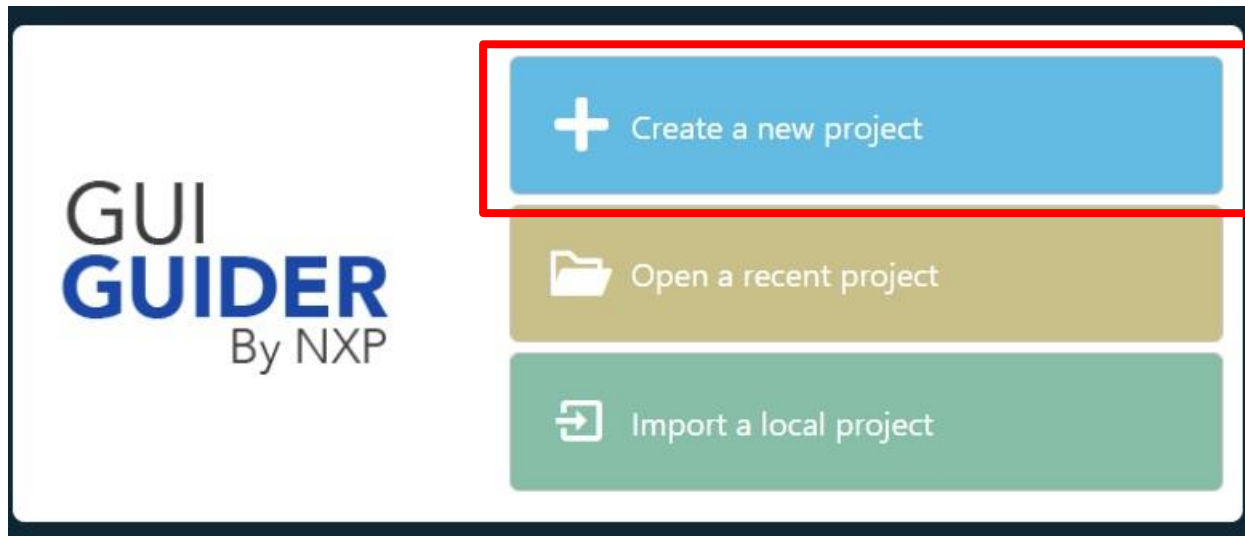
SIGN IN

[Employee Sign In](#)

/ Step 1.

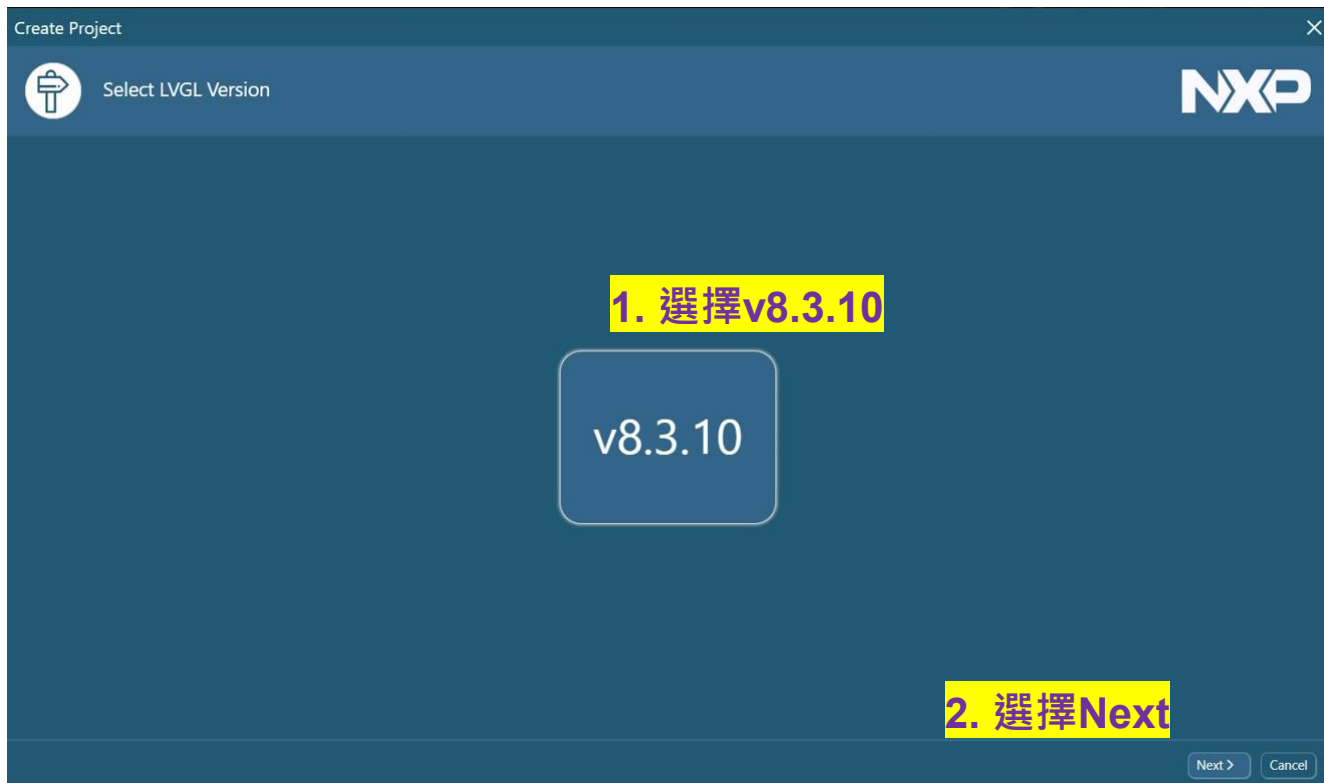


1. 點擊桌面GUI
Guider 開啟

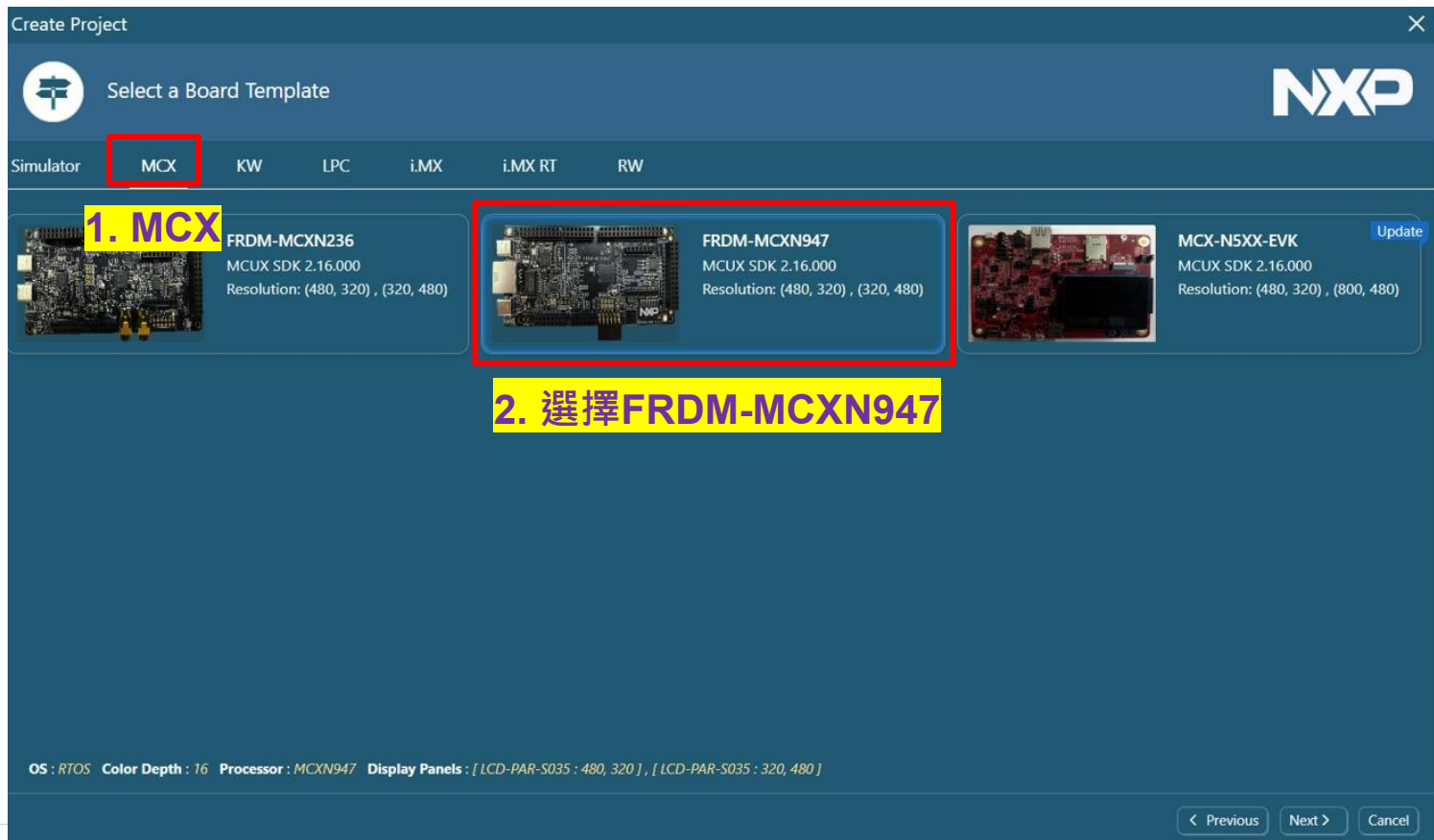


2. 選擇 Create a new project

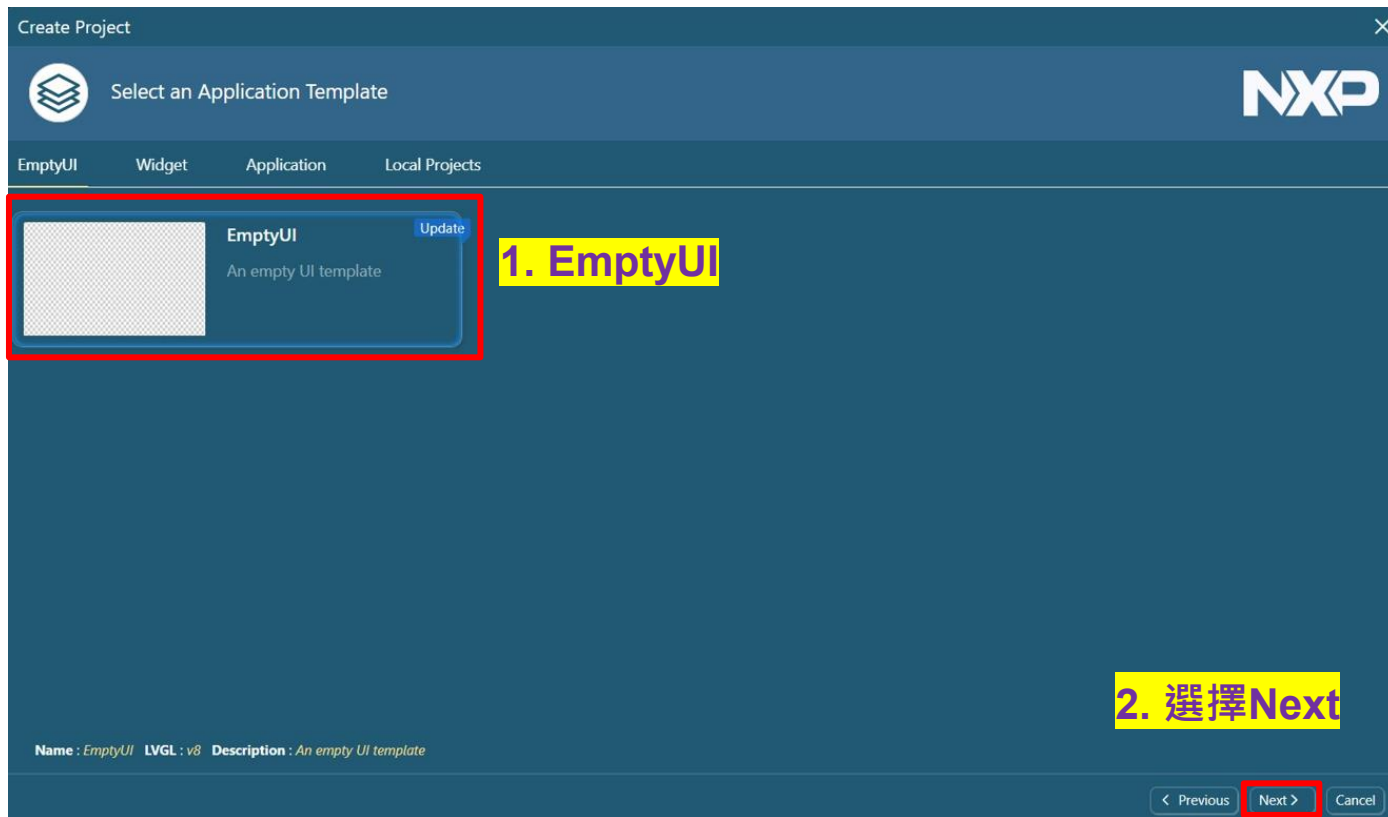
/ Step 2.



/ Step 3.



/ Step 4.

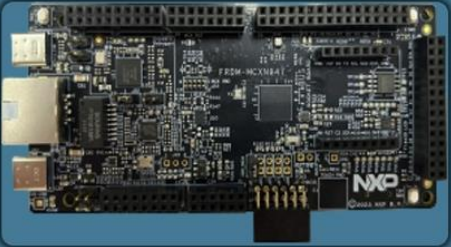


Step 5.

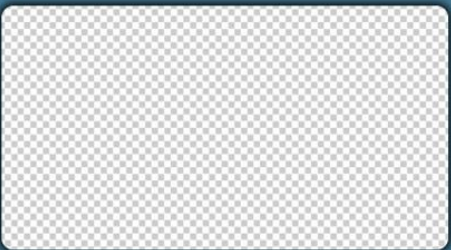
Create Project

Project Settings

NXP



FRDM-MCXX947
MCUX SDK 2.16.000



EmptyUI
LVGL Version: v8
An empty UI template

Project Name

Project Directory

Color Depth
16bit

Panel Type

Zoom
Auto Ratio

1. 鍵入專案名稱

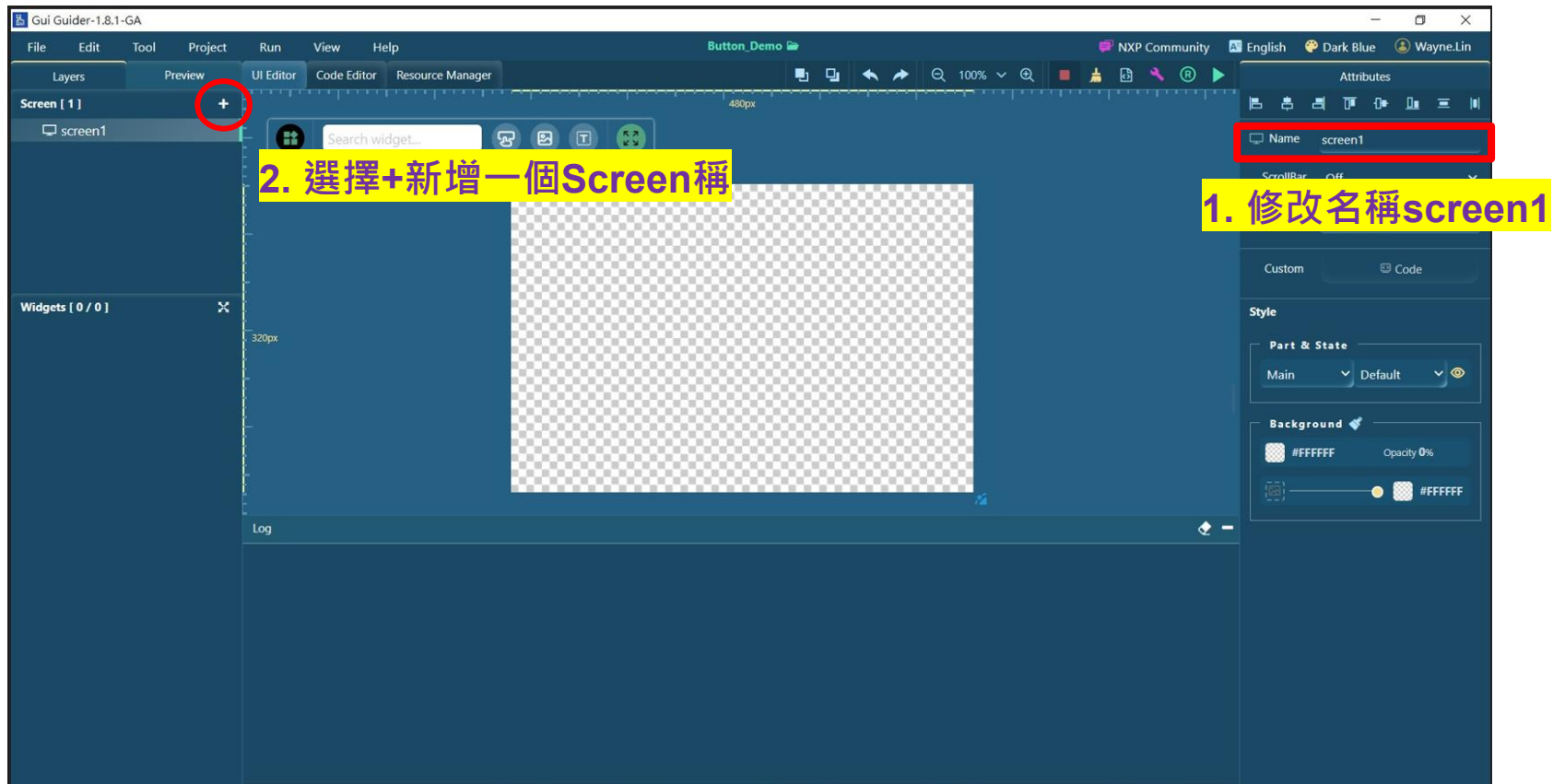
2. 設定專案存檔目錄

3. 選擇Panel型號
LCD-PAR-S035

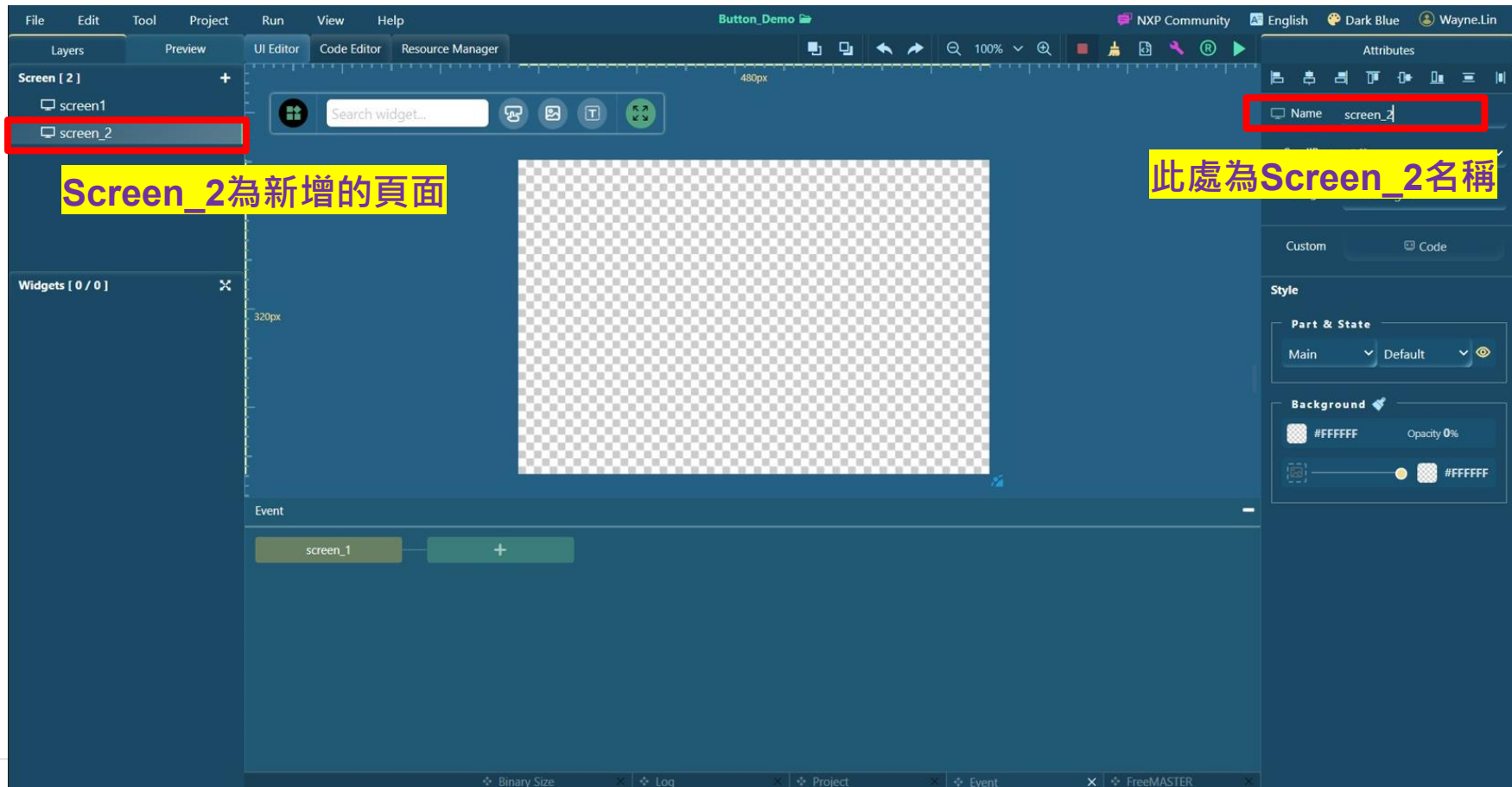
4. 選擇Create建立專案

< Previous Create Cancel

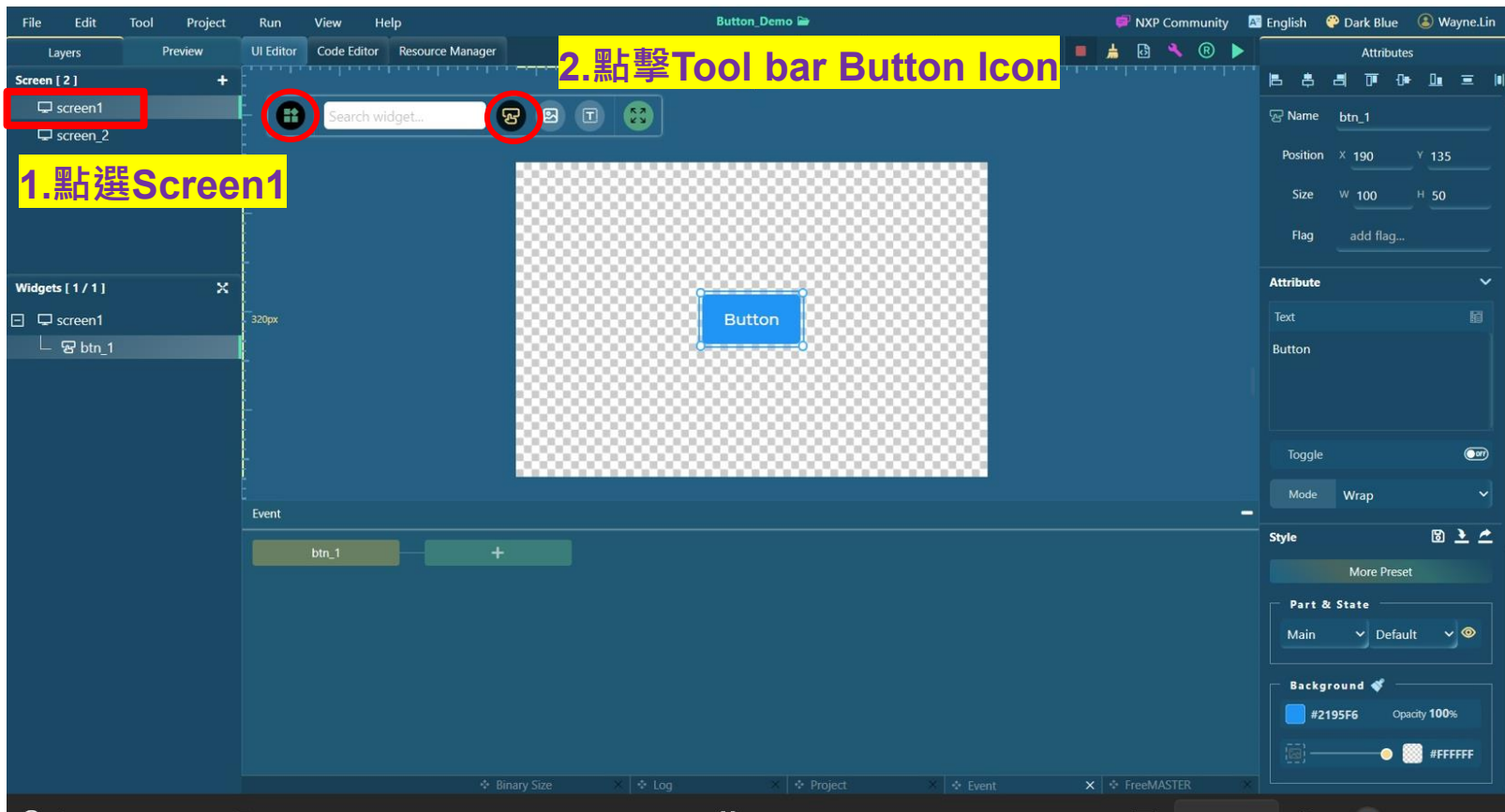
Step 6. 建立一個新的Screen



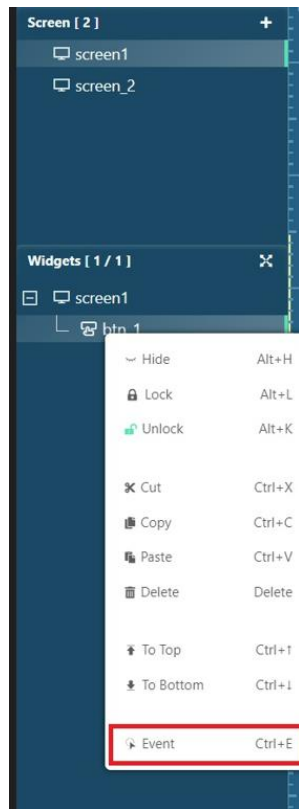
Step 6-1. 建立一個新的Screen



Step 7. 建立一個Button Icon



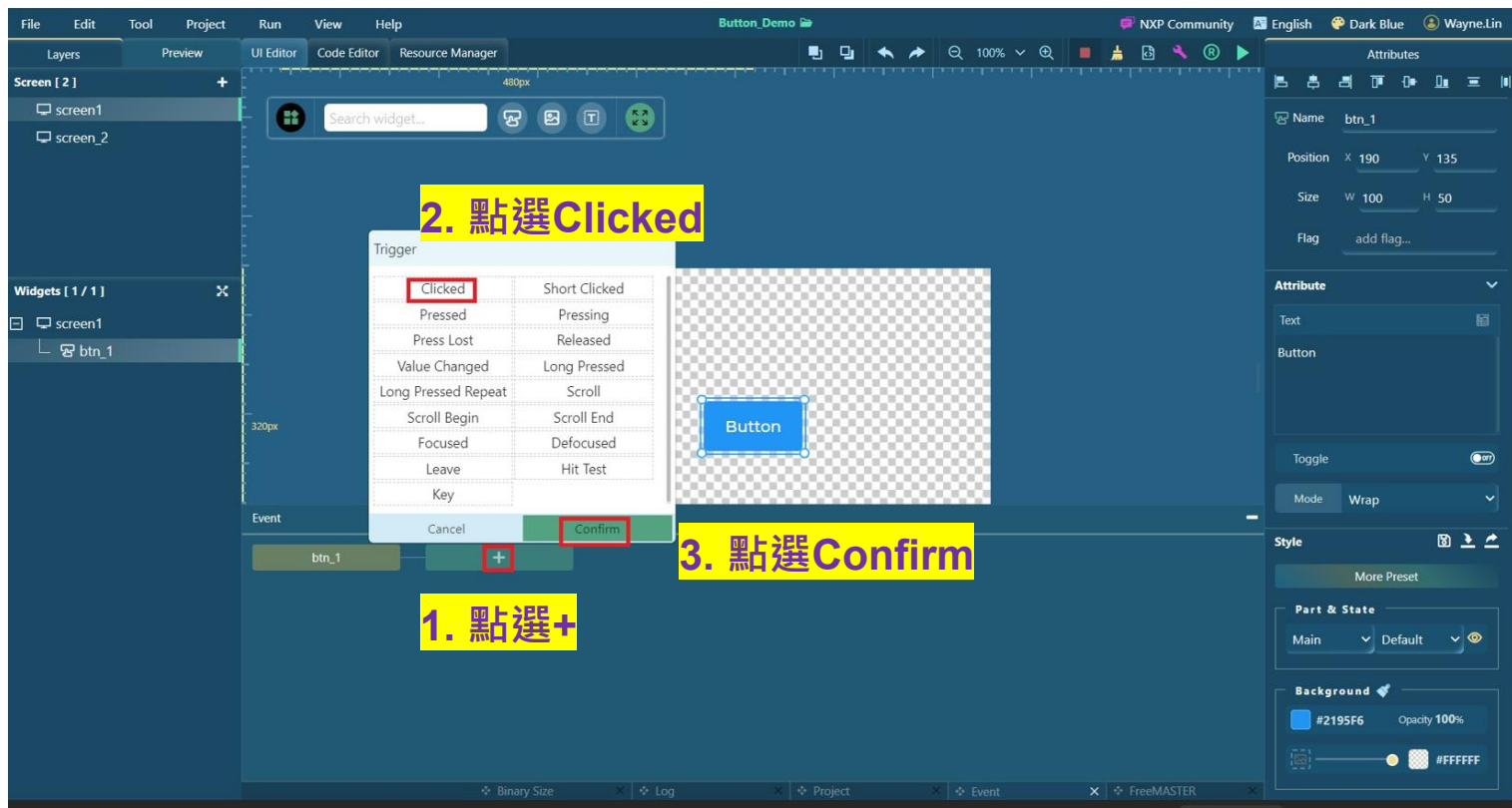
Step 7-1. 建立一個Button Icon



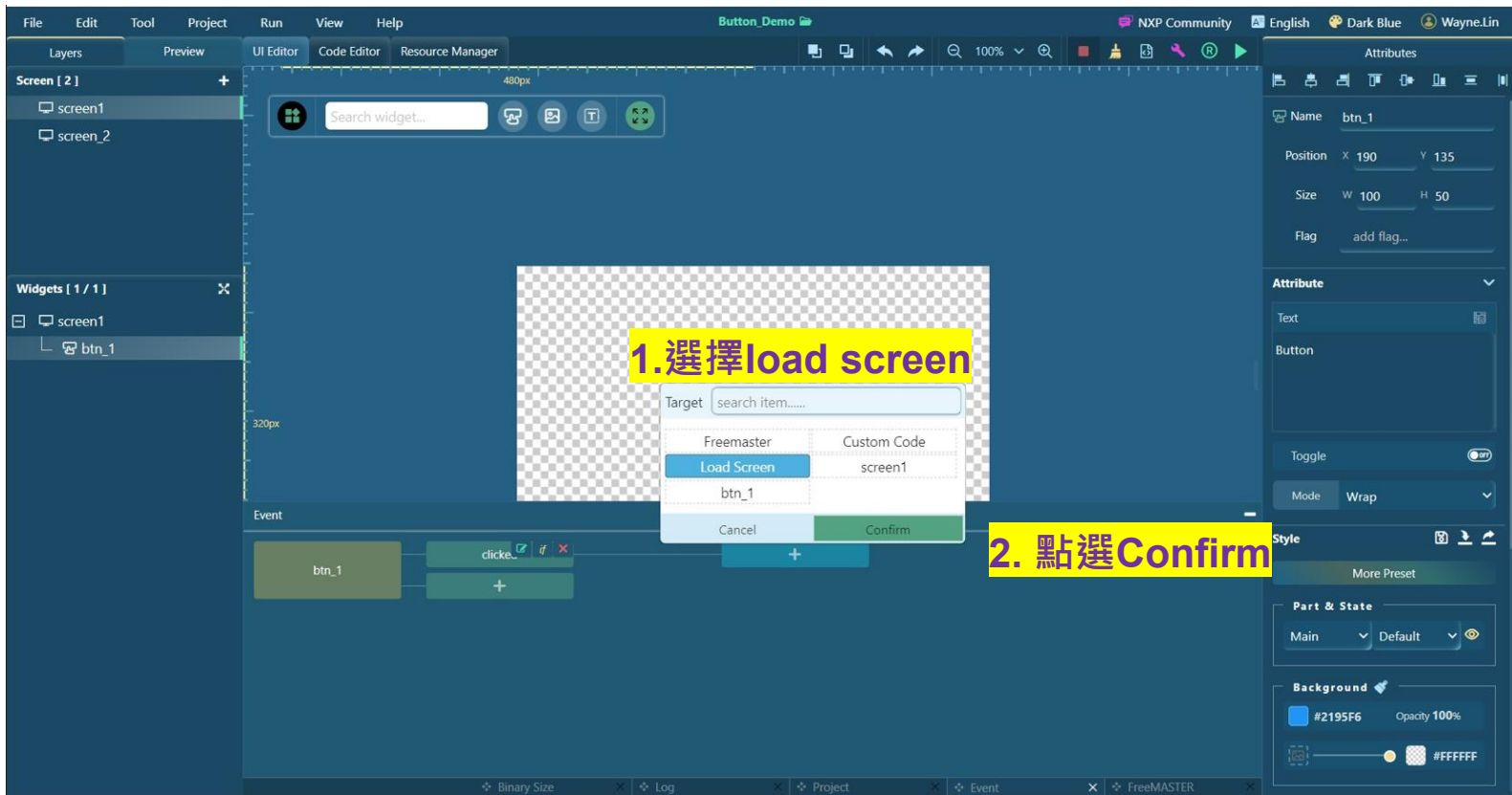
1. 右鍵 btn_1

2. 點擊 Event

Step 7-2. 建立一個Button Icon



Step 7-3. 建立一個Button Icon



Step 7-4. 建立一個Button Icon

The screenshot shows the NXP iMX8M Mini IDE interface. The main canvas displays a button widget labeled "Button" on a transparent background. The "Attributes" panel on the right shows the button's properties: Name: btn_1, Position: X 190, Y 135, Size: W 100, H 50, Flag: add flag... The "Widgets" panel on the left shows the hierarchy: screen1 > btn_1. A yellow text box in the center of the canvas reads: "此設定為選擇點擊此一個Button Icon的動作為轉換畫面至Screen_2". A "Load Screen" dialog box is open, showing the configuration for loading screen_2. The dialog has fields for "screen_2", "animation type" (set to none), "duration" (200), and "delay" (200). There are checkboxes for "Free memory of current screen before loading new screen" (unchecked) and "Delete current screen" (checked). The "Confirm" button is highlighted. A yellow text box next to the dialog reads: "2. Target screen 選擇Screen_2". Below the dialog, the "Event" panel shows a sequence of actions: btn_1 clicked -> if -> load screen -> load_screen -> none. A yellow text box next to the "load_screen" action reads: "1. 選擇none".

此設定為選擇點擊此一個Button Icon的動作為轉換畫面至Screen_2

2. Target screen 選擇Screen_2

1. 選擇none

Step 8. 第二個Screen建立一個"Return" Button

1. 點選Screen_2

2. 點擊Tool bar Button Icon

Step 8-1. 第二個Screen建立一個Button

1. 點選Icon

2. 修改顯示名稱為Return

3. 修改Icon顏色

Step 8-2. 第二個Screen建立一個Button

The screenshot shows the NXP Community IDE interface for creating a button in a second screen. The interface includes a Layers panel, a Widgets panel, a Trigger menu, and a Properties panel.

1. 點選+ (Click +): The button widget **btn_1** is selected in the Widgets panel.

2. 點選Clicked (Click Clicked): The **Clicked** trigger is selected in the Trigger menu.

3. 點選Confirm (Click Confirm): The **Confirm** action is selected in the Event panel.

The button widget **btn_1** is shown in the design area with the text **Return**. The Properties panel on the right shows the button's attributes, including Name (**btn_1**), Position (X: 192, Y: 135), Size (W: 100, H: 50), and Flag (add flag...).

Step 8-3. 第二個Screen建立一個Button

The screenshot shows the NXP Community IDE interface. The main workspace displays a canvas with a checkerboard background. A button labeled "Return" is positioned in the center. A context menu is open over the button, showing options for "Freemaster" and "Custom Code". The "Custom Code" option is selected, and a sub-menu is visible with "Load Screen" and "screen_2". The "Load Screen" option is highlighted. A yellow box with the text "1. 選擇load screen" is overlaid on the "Load Screen" option. Another yellow box with the text "2. 點選Confirm" is overlaid on the "Confirm" button in the context menu. The right sidebar shows the "Attributes" panel for the selected button, with fields for Name, Position, Size, and Flag. The "Attribute" panel shows the "Text" attribute set to "Return". The "Style" panel shows the "Background" attribute set to "#FF6500" with "Opacity 100%".

1. 選擇load screen

2. 點選Confirm

Step 8-4. 第二個Screen建立一個Button

The screenshot shows the NXP Community IDE interface with a project named "Button_Demo". The main canvas displays a blue background with a white grid. A red button labeled "Return" is positioned in the center. A yellow text box with black text is overlaid on the button, stating: "此設定為選擇點擊此一個Return Icon的動作為轉換畫面至Screen1".

On the left side, the "Layers" panel shows "Screen [2]" with "screen1" and "screen_2". The "Widgets [1 / 2]" panel shows "screen_2" and "btn_1".

On the right side, the "Attributes" panel shows the properties for "btn_1": Name: btn_1, Position: X 192, Y 135, Size: W 100, H 50, Flag: add flag... The "Attribute" dropdown is set to "Attribute".

At the bottom, the "Event" panel shows a sequence of events: "btn_1" (clicked) leads to "load screen" (load_screen: none). A yellow text box with black text is overlaid on the "load screen" event, stating: "1. 選擇none".

A "Load Screen" dialog box is open, showing the configuration for loading "screen1". The dialog has fields for "screen1", "duration" (200), and "delay" (200). There are checkboxes for "Free memory of current screen before loading new screen" (unchecked) and "Delete current screen" (checked). The "Confirm" button is highlighted. A yellow text box with black text is overlaid on the dialog, stating: "2. Target screen 選擇Screen1".

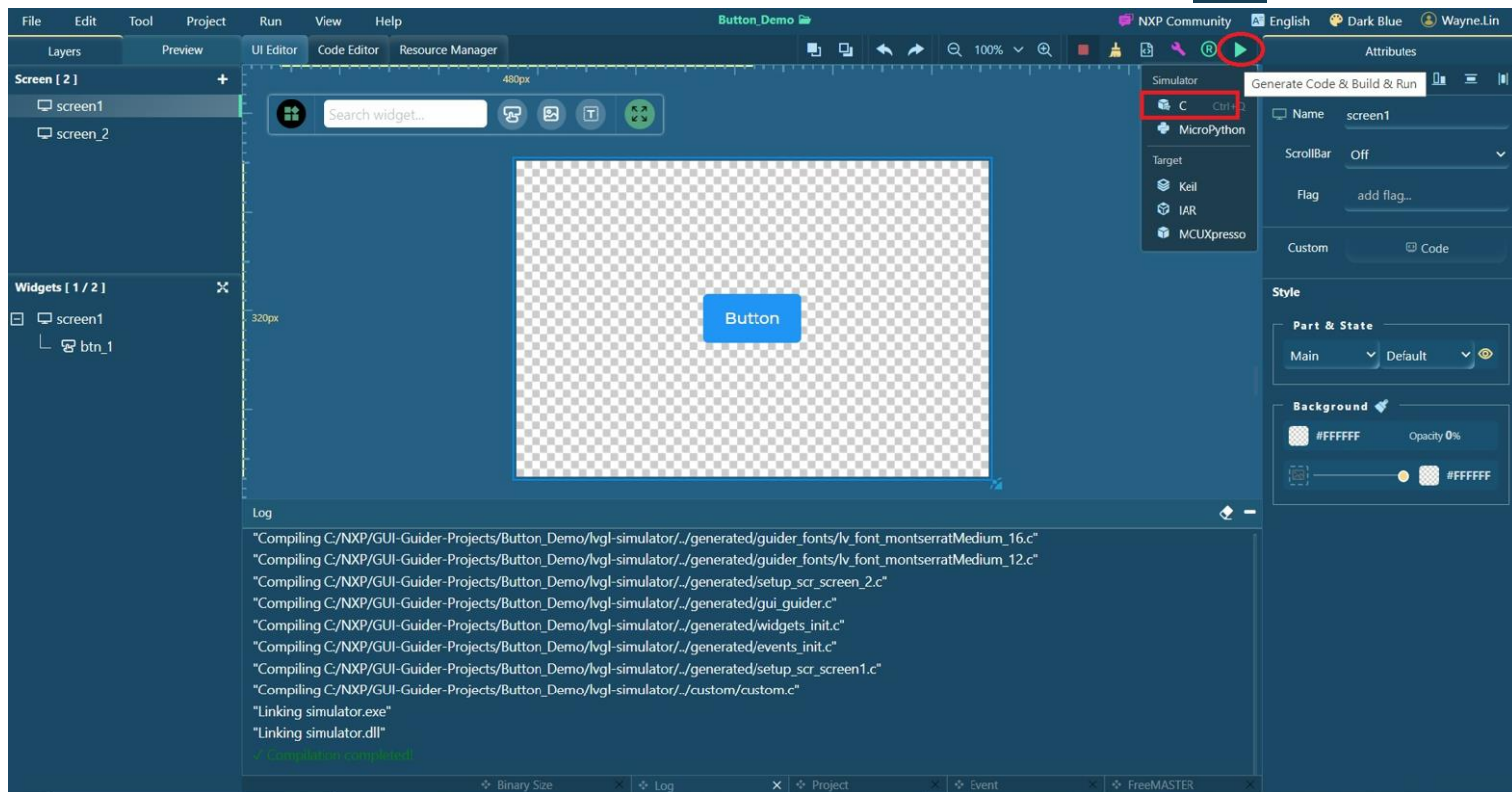
Another yellow text box with black text is overlaid on the "Confirm" button, stating: "3. 點選Confirm".

Step 9. PC執行模擬

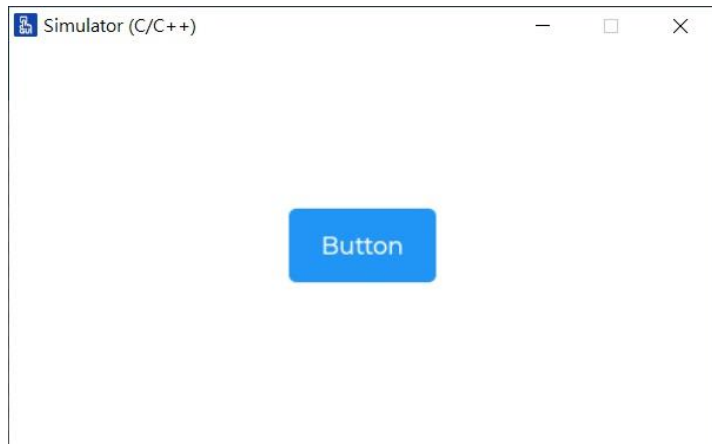
選擇



符號並選擇 Simulator → C



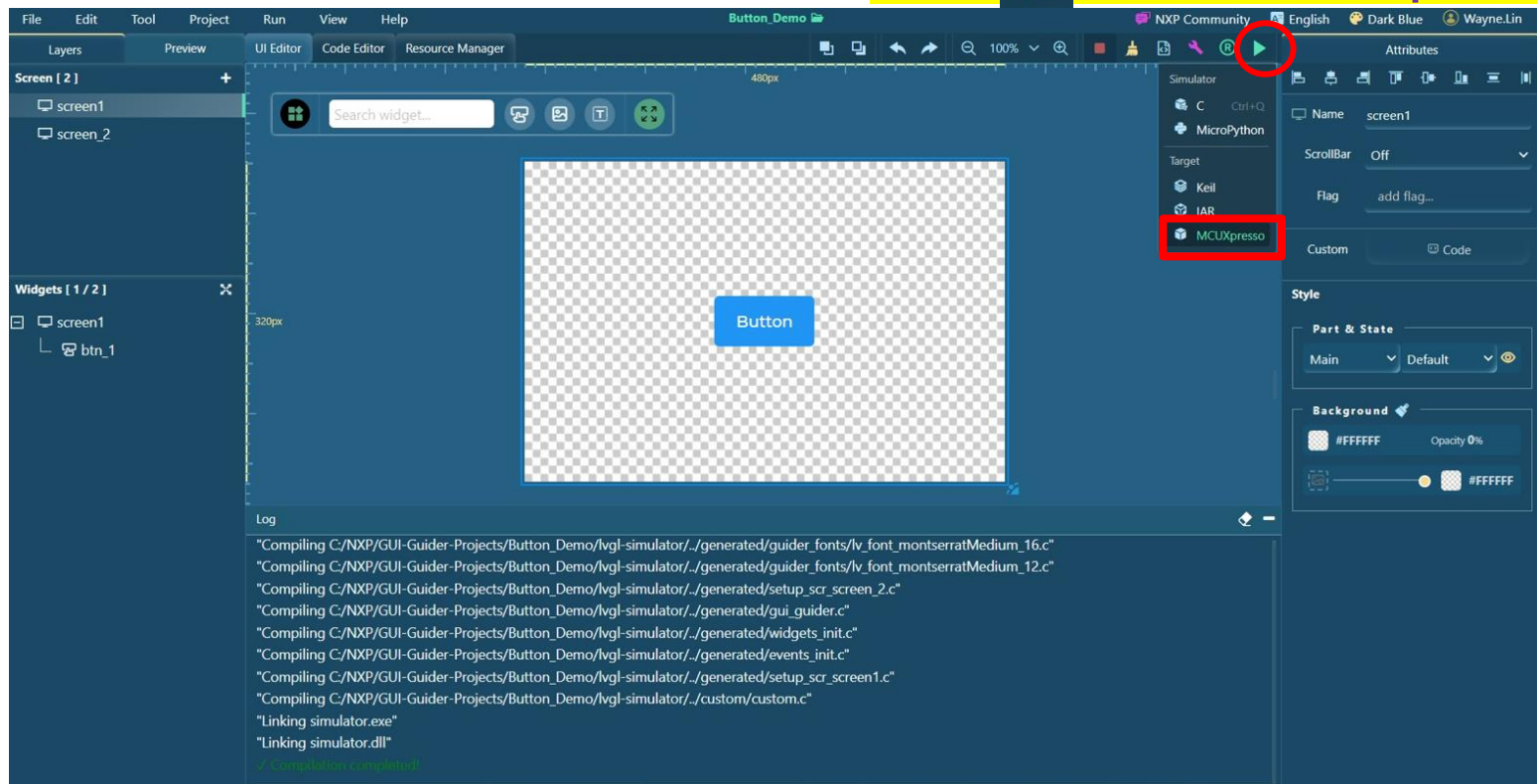
/ Step 9-1. PC執行模擬



Step 10. 載入FRDM-MCXXN947

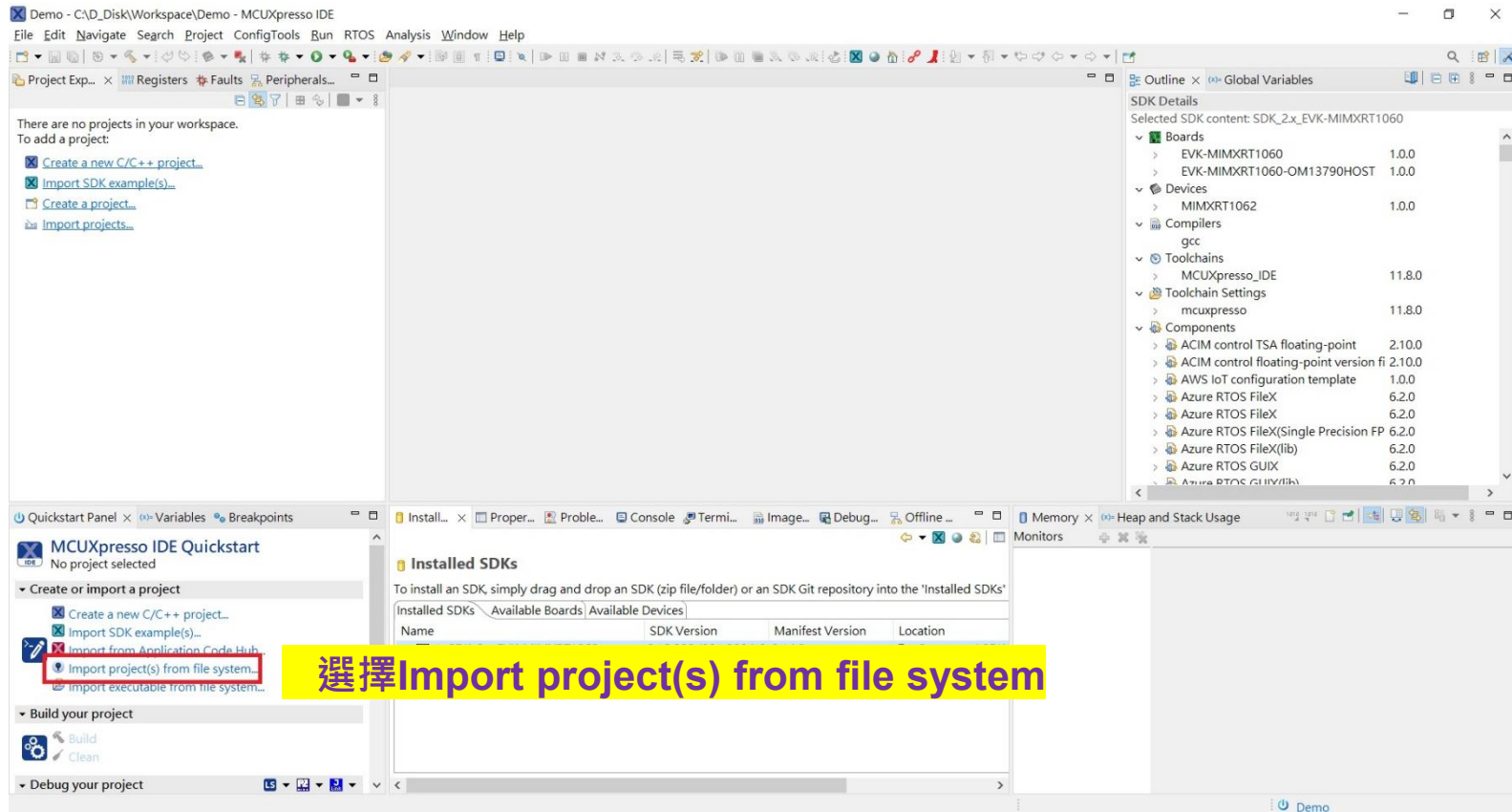
選擇

符號並選擇MCUXpresso



Import GUI Guider project to MCUXpresso

Step 1. Import GUI Project



Step 1-1. Import GUI Project

Import project(s) from file system...

Import project(s) from file system...

Select the examples archive file to import.

Projects are contained within archives (.zip) or are unpacked within a directory. Select your project archive or root directory and press <Next>. On the next page, select those projects you wish to import, and press <Finish>.

Project archives for LPCOpen and 'legacy' examples are provided.

Project archive (zip)

Archive

Project directory (unpacked)

Root directory C:\NXP\GUI-Guider-Projects\Button_Demo

Browse...

LPCOpen

LPCOpen is the recommended software for LPC parts introduced before 2016. It is not recommended for new LPC8xx developments. MCUXpresso IDE includes the LPCOpen packages which can be imported directly by pressing the Browse button in the Project archive (zip) section, above, and navigating to the Examples/LPCOpen directory.

Alternatively, press the button below to Browse the nxp.com website for latest resources.

Browse LPCOpen resources on nxp.com...

4. 確認為 Button_Demo www.nxp.com/pages/LPC-OPEN-LIBRARIES

< Back

Next >

Finish

Cancel

Select the root directory containing the projects

<< NXP > GUI-Guider-Projects > Button_Demo >

搜尋 Button_Demo

組合管理 新增資料夾

名稱	修改日期	類型	大小
custom	2024/3/12 下午 06:40	檔案資料夾	
generated	2024/3/12 下午 07:20	檔案資料夾	
import	2024/3/12 下午 06:40	檔案資料夾	
lib	2024/3/12 下午 06:40	檔案資料夾	
lvgl	2024/3/12 下午 06:40	檔案資料夾	
lvgl-simulator	2024/3/12 下午 07:01	檔案資料夾	
ports	2024/3/12 下午 06:40	檔案資料夾	
sdk	2024/3/12 下午 06:40	檔案資料夾	
temp	2024/3/12 下午 07:01	檔案資料夾	

2. 確認為 Button_Demo

資料夾: Button_Demo

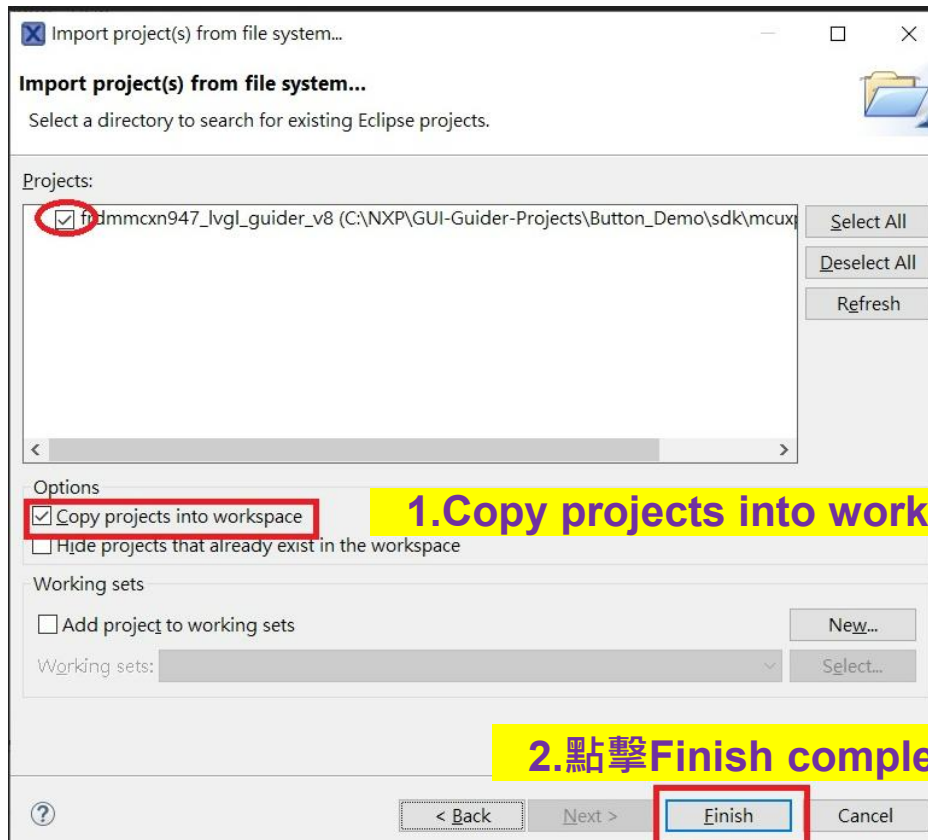
選擇資料夾

取消

3. 開啟選擇資料夾

5. 開啟選擇資料夾

Step 1-2. Import GUI Project



1. Copy projects into workspace

2. 點擊 Finish complete import GUI project

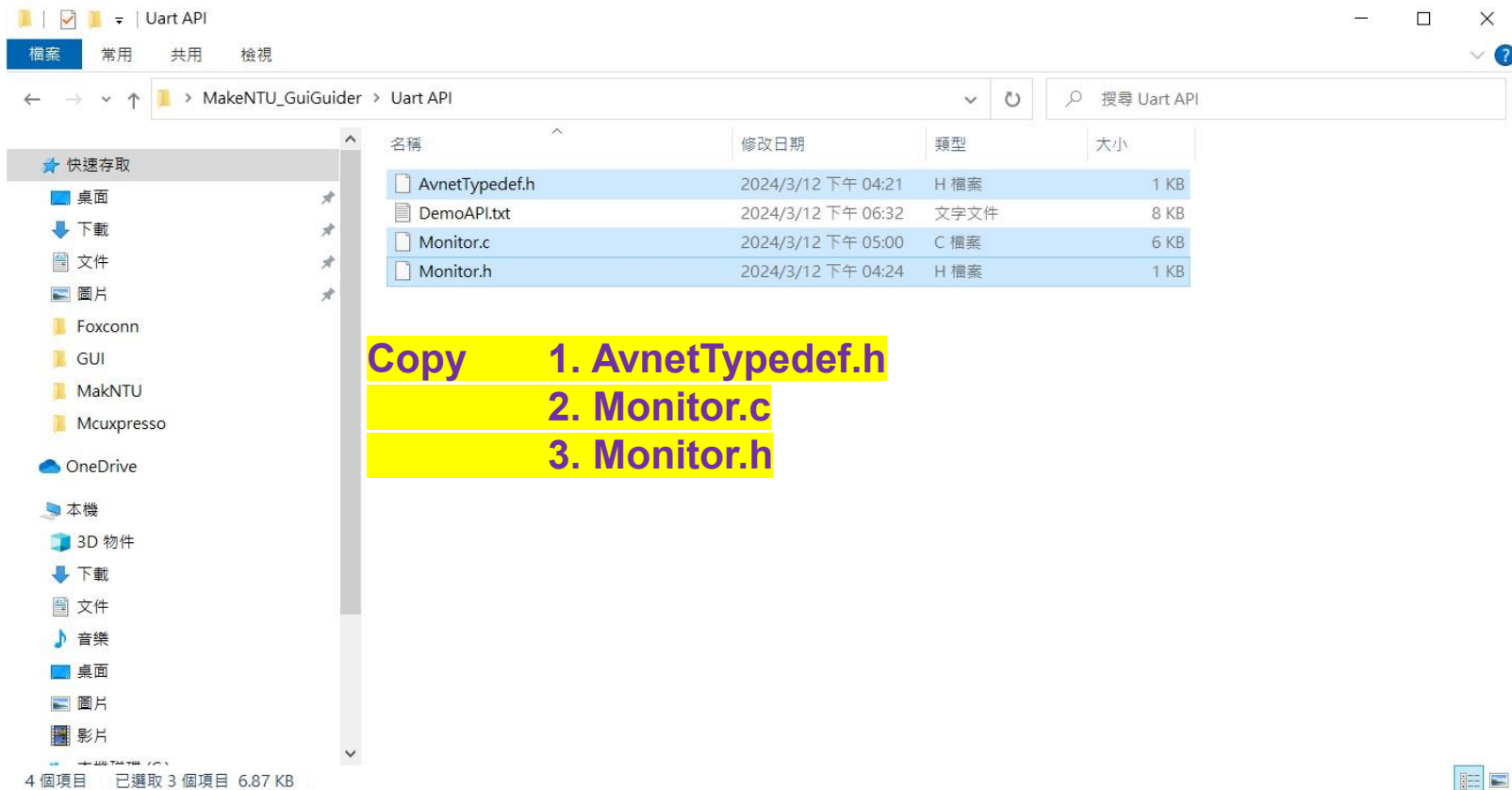
Step 2. Import Project complete

The screenshot shows the MCUXpresso IDE interface. On the left, the Project Explorer displays a tree structure for the project 'evkmimxrt1060_lvgl_guide_v8'. A yellow text box with purple text is overlaid on the Project Explorer, stating: 'Import 完 project 之後可以在左側畫面看到完整的樹狀專案' (After importing the project, you can see the complete tree structure in the left panel). The bottom-left pane shows the 'MCUXpresso IDE Quickstart' panel with options to create or import a project. The bottom-right pane shows the 'Installed SDKs' table.

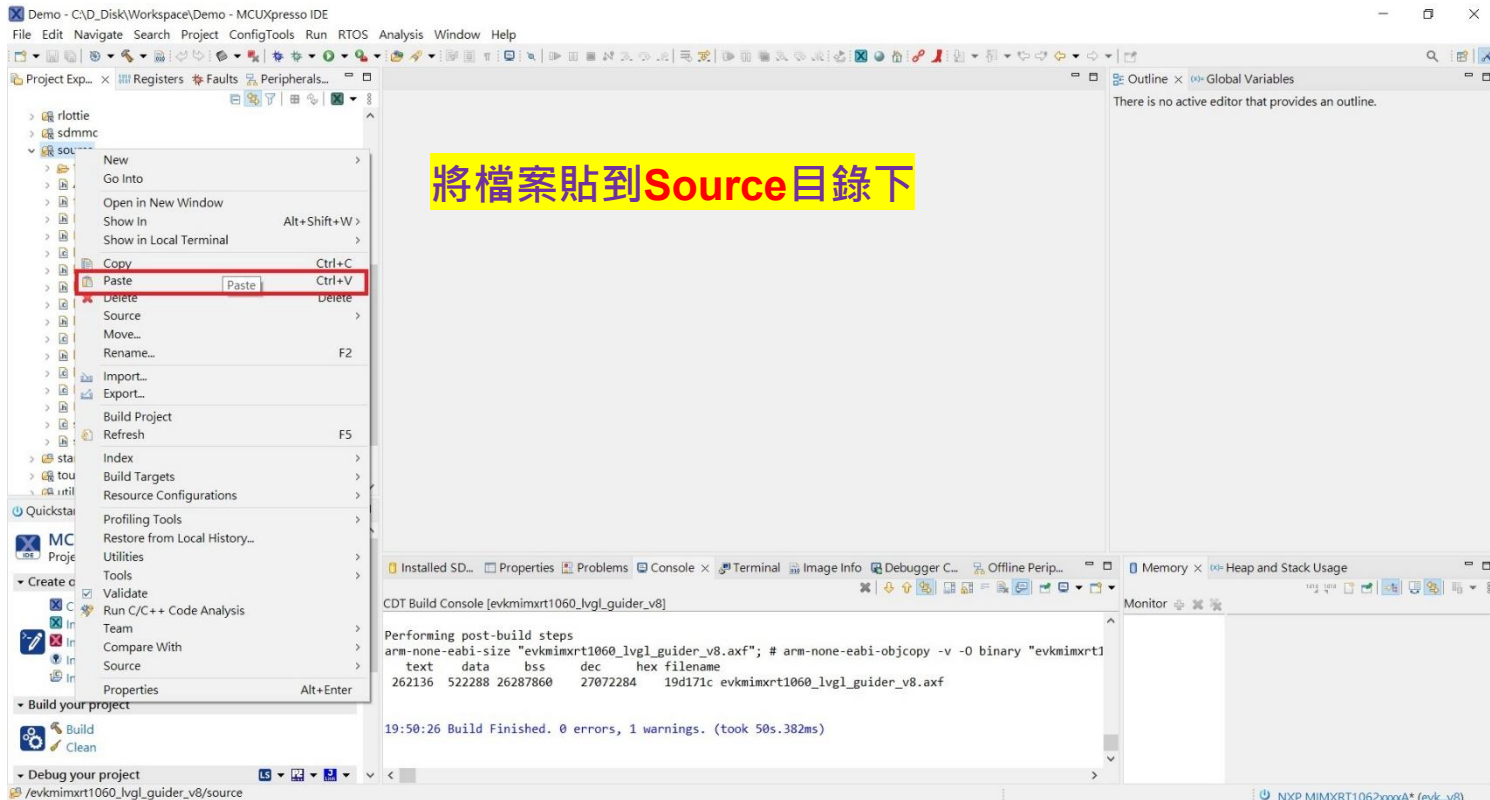
Name	SDK Version	Manifest Version	Location
☒ SDK_2x-EVK-MIMXRT1060	2.15.000 (801 2024-0	3.14.0	<Common>\SDK_2_15_000-EVK-MIMXR

Implement Demo API

Step 1. Copy Demo code to MCUxpresso



Step 2. Copy Demo code to MCUxpresso



Step 4. Copy function into Project

AvnetTypeDef.h

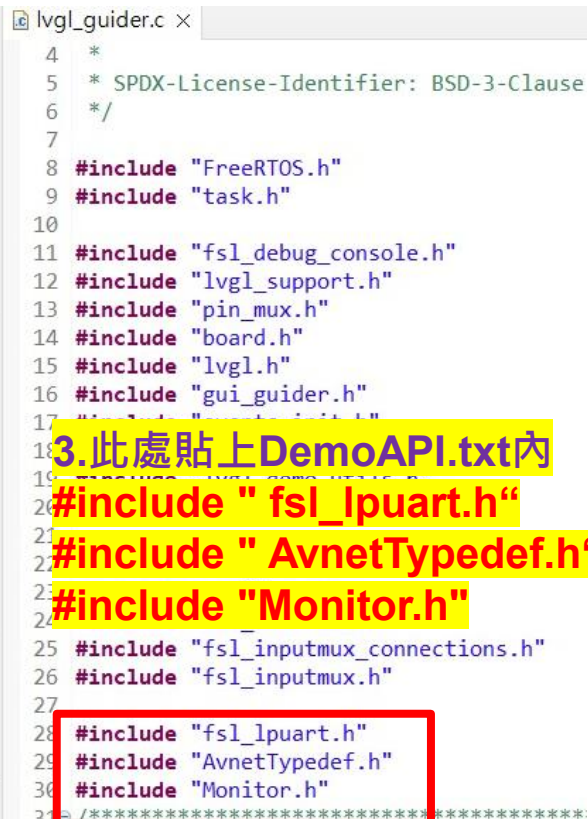
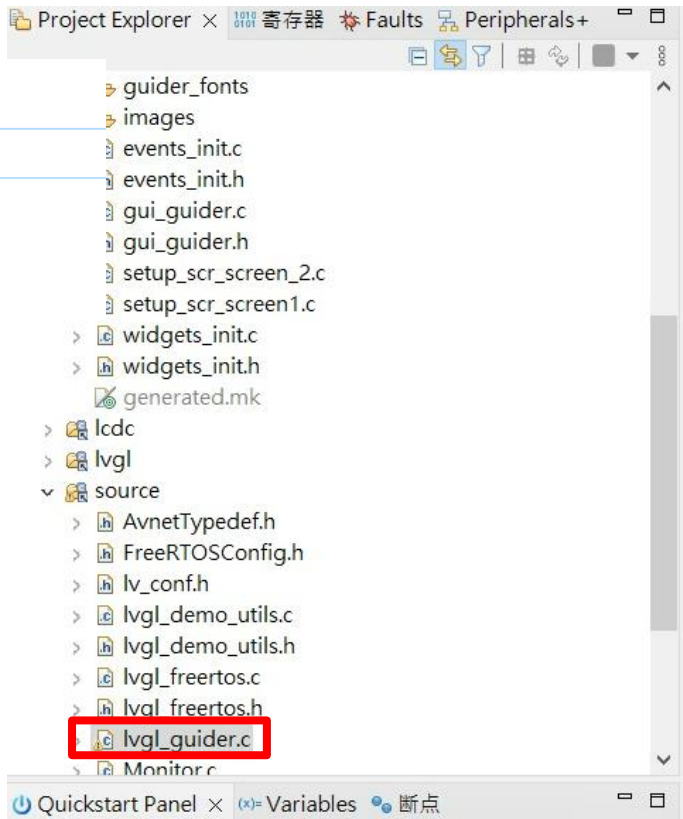
DemoAPI.txt

Monitor.c

Monitor.h

1. 使用記事本開啟 DemoAPI

2. 選擇 lvgl_guider.c



3. 此處貼上 DemoAPI.txt 內

#include "fsl_lpuart.h"
#include "AvnetTypeDef.h"
#include "Monitor.h"

Step 4. Copy function into Project

```

98= /*****
99  * Code
100 *****/
101
102= /*!
103  * @brief Main function
104  */
105

```

1. 將API Function 拷貝至程式main 之上

```

106+ void InitUartConnect(){}
123
124+ void ButtonGui(char* pValue){}
128
129+ void ReturnGui(char* pValue){}
133
134+ void ShowNumber(char* pValue){}
141
142
143 SMONITORCOMMAND sMonitorFuncList[]=
144 {
145     { "Button",      "<var 1> <var 2>",      "Print the input numbers",      ButtonGui },
146     { "Return",      "<var 1> <var 2>",      "Print the input numbers",      ReturnGui },
147     { "ShowNum",      "<var 1> <var 2>",      "Print the input numbers",      ShowNumber },
148     {0,0,0,0}

```

2. 請注意程式要拷貝完整,此處程式因截圖關係已經縮排

/ Step 4. Copy function into Project

```
173 void InitUartConnect()  
174 {  
175     lpuart_config_t config;  
176  
177     LPUART_GetDefaultConfig(&config);  
178     config.baudRate_Bps = BOARD_DEBUG_UART_BAUDRATE;  
179     config.enableTx      = true;  
180     config.enableRx      = true;  
181     LPUART_Init(DEMO_LPUART, &config, DEMO_LPUART_CLK_FREQ);  
182     MonitorInit();  
183  
184     /* Enable RX interrupt. */  
185     LPUART_EnableInterrupts(DEMO_LPUART, kLPUART_RxDataRegFullInterruptEnable);  
186     EnableIRQ(DEMO_LPUART_IRQn);  
187 }
```

如前頁所述請務必完整拷貝程式內容

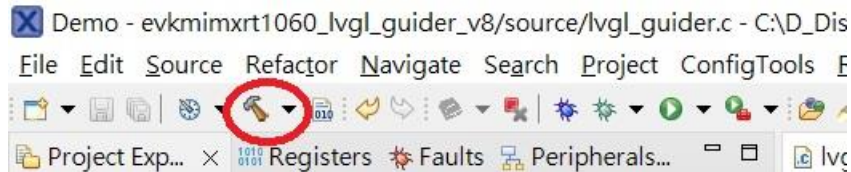
Step 5. Copy Initial function into Project

```
lvgl_guider.c x
212  * @brief Main function
213  */
214  int main(void)
215  {
216      BaseType_t stat;
217
218      /* Init board hardware. */
219      /* attach FRO 12M to FLEXCOMM4 (debug console) */
220      CLOCK_SetClkDiv(kCLOCK_DivFlexcom4Clk, 1u);
221      CLOCK_AttachClk(BOARD_DEBUG_UART_CLK_ATTACH);
222
223      /* attach FRO 12M to FLEXCOMM2 */
224      CLOCK_SetClkDiv(kCLOCK_DivFlexcom2Clk, 1u);
225      CLOCK_AttachClk(kFRO12M_to_FLEXCOMM2);
226
227      CLOCK_SetClkDiv(kCLOCK_DivFlexioClk, 1u);
228      CLOCK_AttachClk(kPLL0_to_FLEXIO);
229      BOARD_InitBootClocks();
230      BOARD_I2C_ReleaseBus();
231      BOARD_InitBootPins();
232      BOARD_InitDebugConsole();
233      DEMO_InitUsTimer();
234
235      /* Init smartdma. */
236      BOARD_InitSmartDMA();
237
238      // Initial Uart
239      InitUartConnect();
240
241      stat = xTaskCreate(AppTask, "lvgl", configMINIMAL_STACK_SIZE + 800, NULL, tskIDLE_PRIORITY + 2, NULL);
```

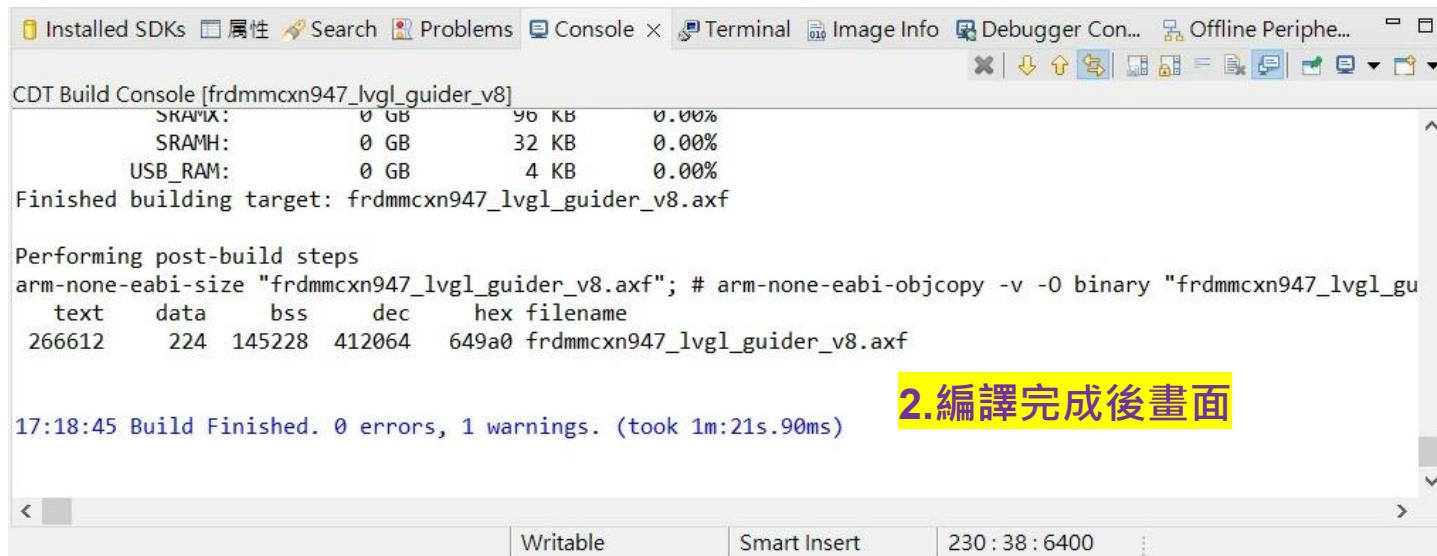
將InitUartConnect() 加入 main

Compile and download

Step 1. Compile

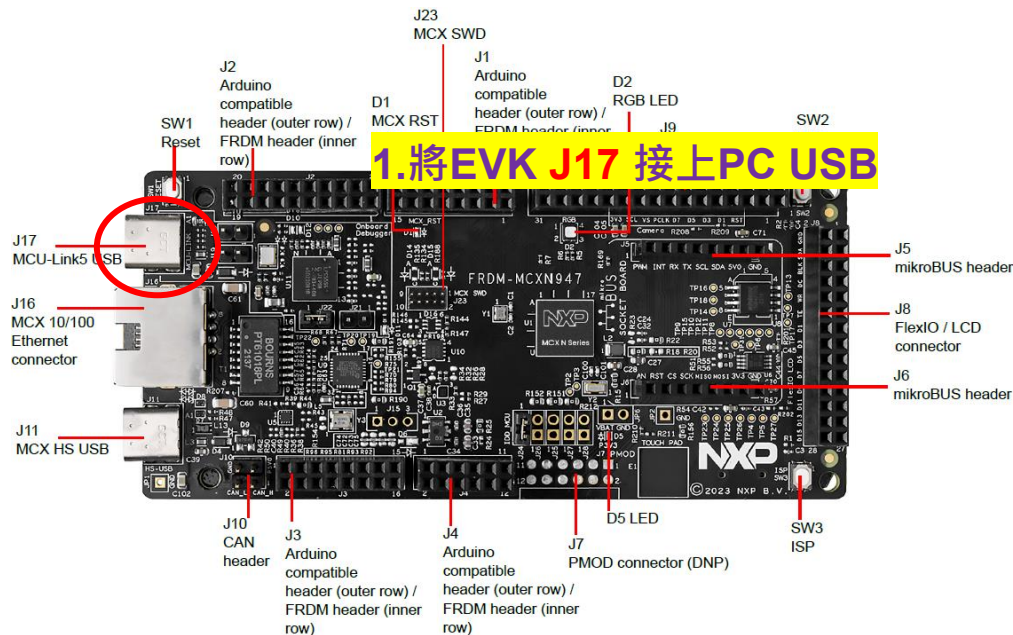


1. 點選槌子符號開始Compile

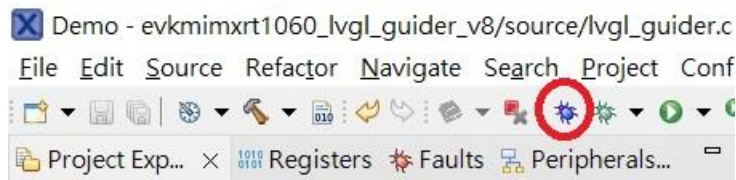


2. 編譯完成後畫面

Step 2. Download



1.將EVK J17 接上PC USB



2.點選藍色蟲子符號開始Download code

Step 2-1. Download

☒ Probes discovered

Connect to target: MCXN947

⚠ Firmware update(s) available for 1 of the discovered probes.
1 probe found. Auto-selected probe matching project's target. Detected board FRDM-MCXN947 and device MCXN947VDF.

Available attached probes

Name	Serial number / ID / Nickn...	Type	Manufacturer	IDE Debug Mode
LS ⚠ MCU-LINK FRDM-MCXN947 (r0E7) CMSIS-DAP V3.128	13QEGEX3JDGK5	LinkServer	NXP Semiconductors	Non-Stop

Supported Probes (tick/untick to enable/disable)

☒ MCUXpresso IDE LinkServer (inc. CMSIS-DAP) probes

☒ P&E Micro probes

☒ SEGGER J-Link probes

Probe search options

☒ Remember my selection (for this Launch configuration)

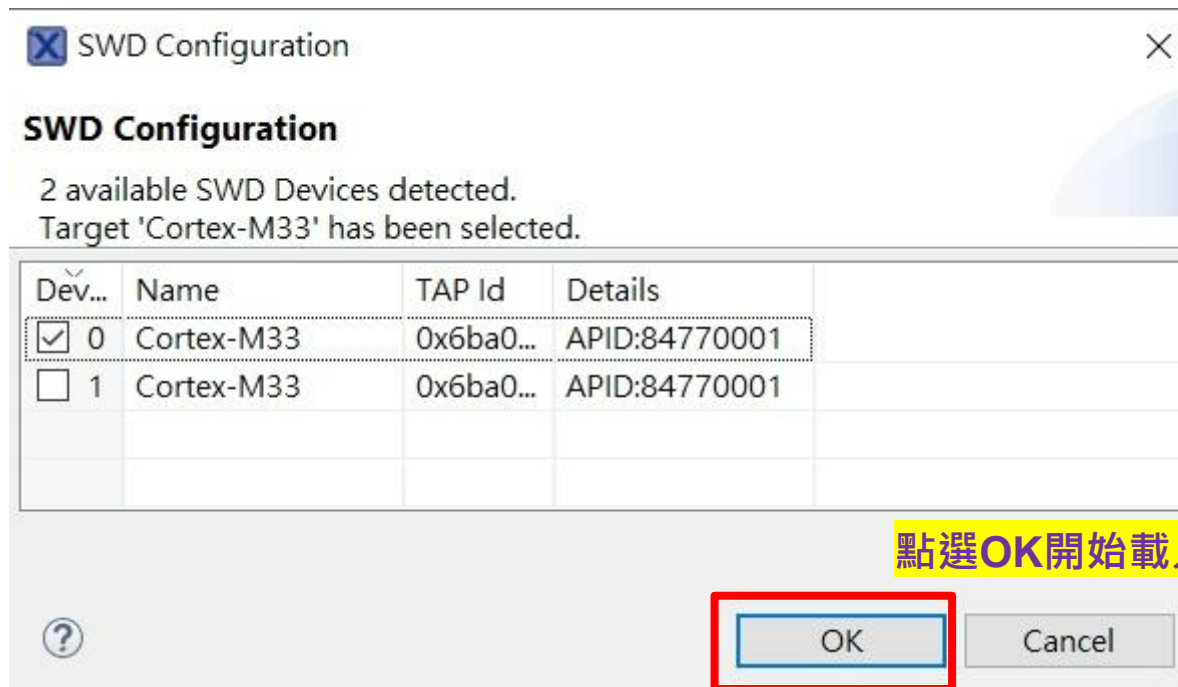
?

OK

Cancel

點選OK開始載入程序

/ Step 2-2. Download

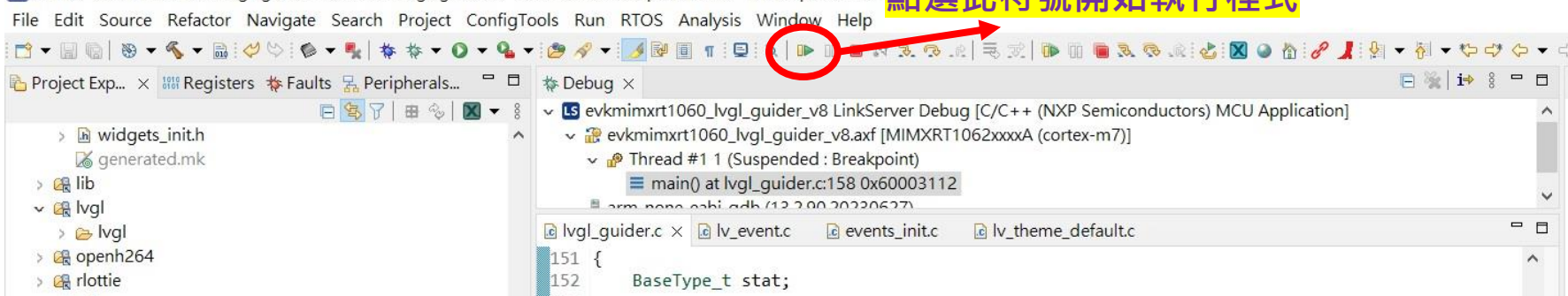


Step 3. Run

載入完成後會進入debug 模式

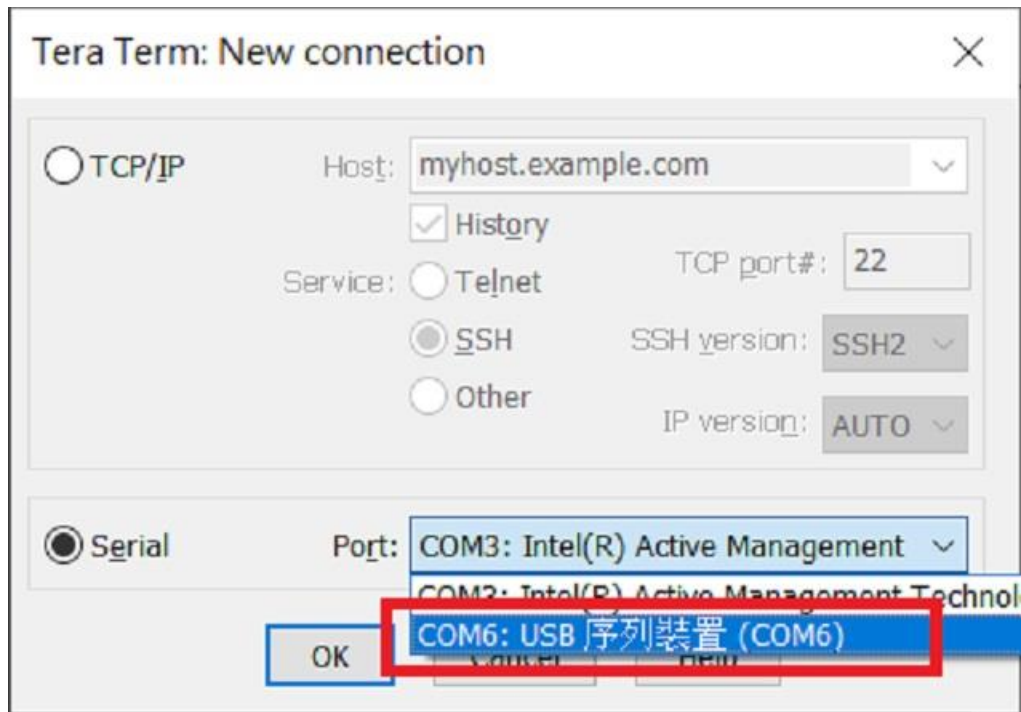
Demo - evkmimxrt1060_lvgl_guider_v8/source/lvgl_guider.c - C:\D_Disk\Workspace\Demo - MCUXpresso IDE

點選此符號開始執行程式



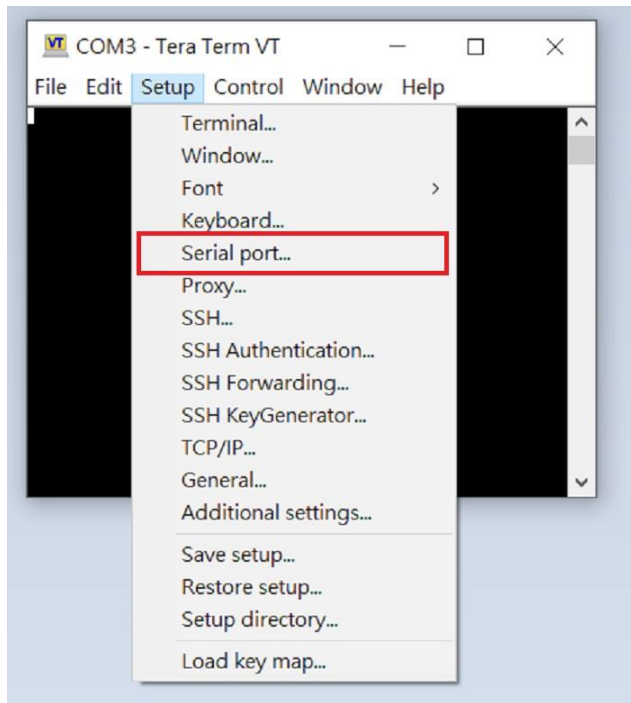
UART Connect

Step 1. 開啟TeraTerm



選擇目前使用COM Port

Step 2. Setting TeraTerm



選擇 Setup → Serial port

Step 2-1. Setting TeraTerm

Tera Term: Serial port setup and connection

Port: COM3

Speed: 115200

Data: 8 bit

Parity: none

Stop bits: 1 bit

Flow control: none

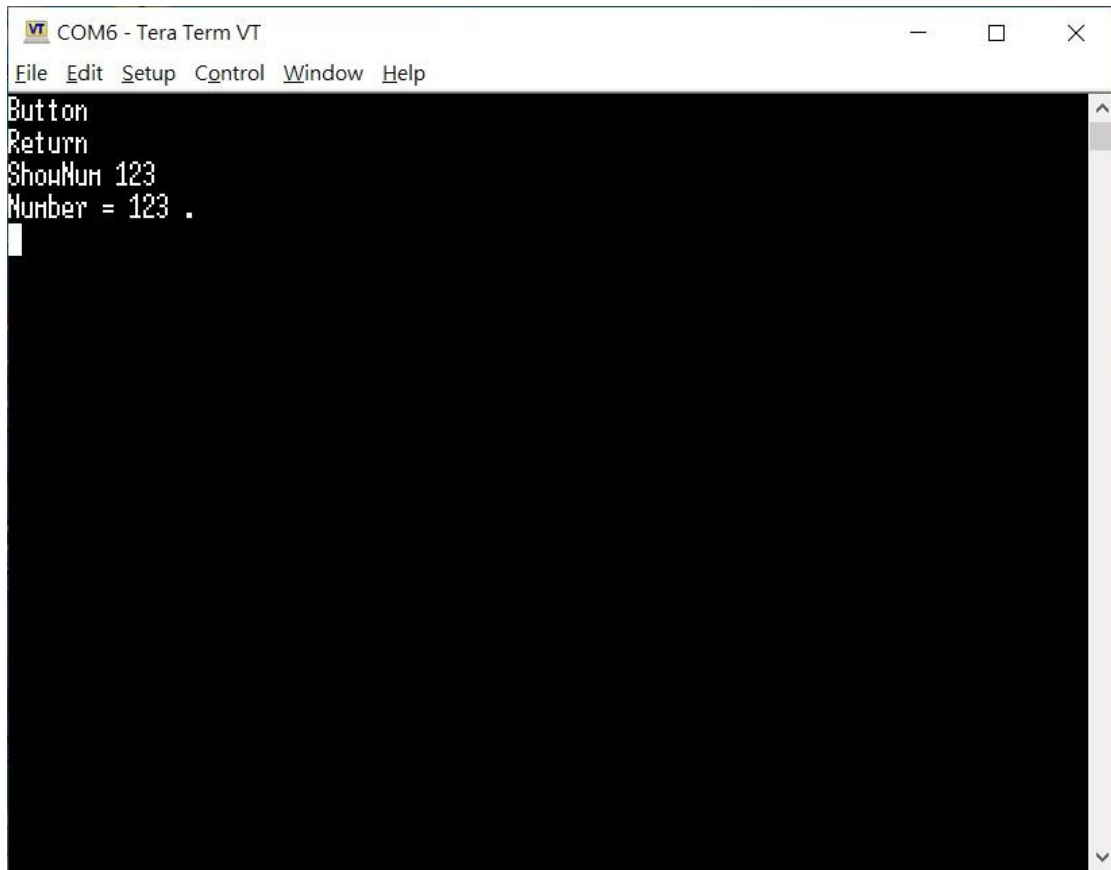
Transmit delay

0 msec/char 0 msec/line

Device Friendly Name: Intel(R) Active Management Technology -
Device Instance ID: PCI\VEN_8086&DEV_9D3D&SUBSYS_504A1;
Device Manufacturer: Intel
Provider Name: Intel
Driver Date: 10-14-2020
Driver Version: 2042.0.13.0

1. Speed 設定為115200
2. 點選New setting完成baud rate設定

Step 3. 鍵入測試命令測試



The screenshot shows a Tera Term VT window titled "COM6 - Tera Term VT". The menu bar includes "File", "Edit", "Setup", "Control", "Window", and "Help". The terminal area has a black background with white text. The text displayed is: "Button", "Return", "ShowNum 123", and "Number = 123 .". A white cursor is positioned at the end of the last line.

```
COM6 - Tera Term VT
File Edit Setup Control Window Help
Button
Return
ShowNum 123
Number = 123 .

```

預設DemoAPI有三個

1. Button
2. Return
3. ShowNum n

GUI Guider Function code

1. Demo API中所呼叫的相關程式放在generated→events_init.c
2. GUI Guider相關程式皆在generated目錄之下。

```
void ButtonGui(char* pValue)
```

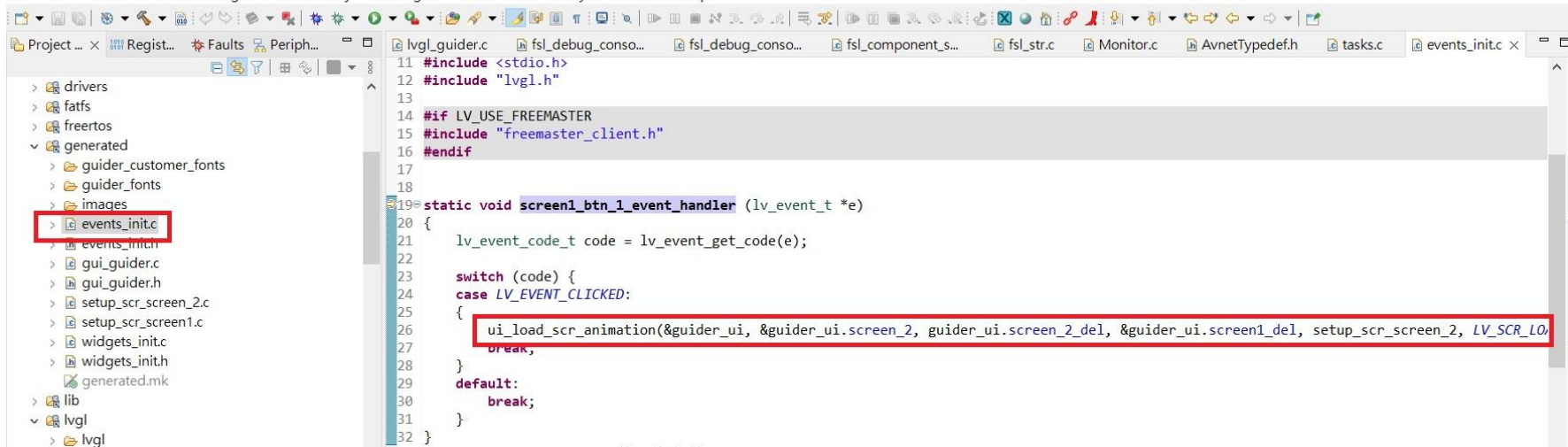
```
{
```

```
    ui_load_scr_animation(&guider_ui, &guider_ui.screen_2, guider_ui.screen_2_del, &guider_ui.screen1_del, setup_scr_screen_2, LV_SCR_LOAD_ANIM_NONE, 200, 200, true, true);
```

```
}
```

☒ Demo - evkmimxrt1060_lvgl_guider_v8/generated/events_init.c - C:\Disk\Workspace\Demo - MCUXpresso IDE

File Edit Source Refactor Navigate Search Project ConfigTools Run RTOS Analysis Window Help



How to add API

Add API

```
void ButtonGui(char* pValue)
{
    ui_load_scr_animation(&guider_ui, &guider_ui.screen_2, guider_ui.screen_2_del, &guider_ui.screen1_del, setup_scr_screen_2, LV_SCR_LOAD_
}

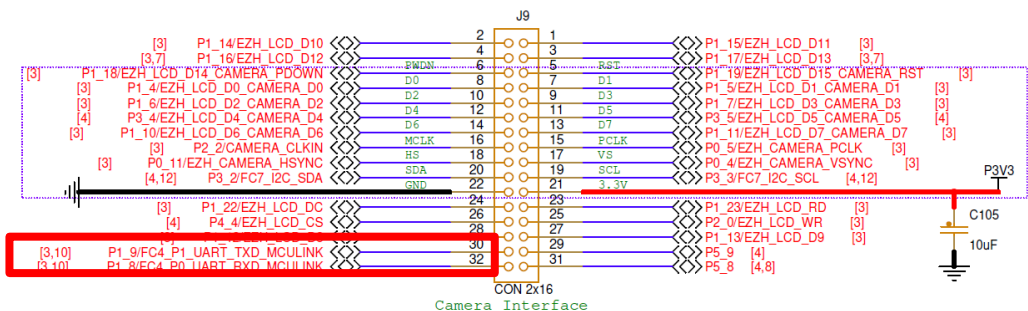
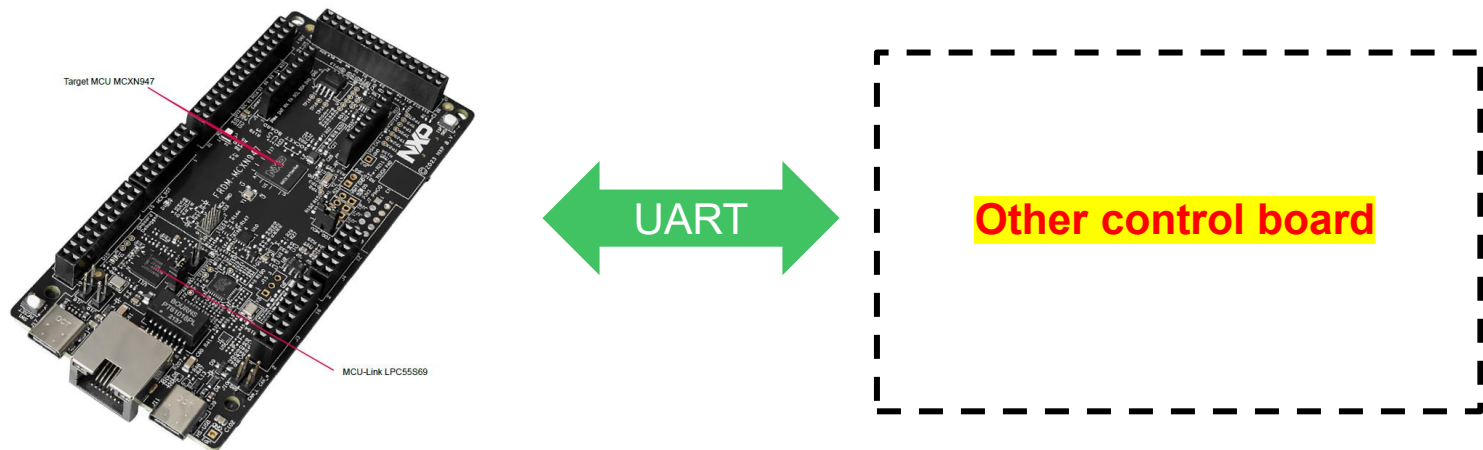
void ShowNumber(char* pValue)
{
    unsigned int value=0;
    sscanf(pValue,"%d",&value);
    PRINTF("Number = %d .\r\n" , value);
}

SMONITORCOMMAND sMonitorFuncList[]=
{
    { "Button", "<var 1> <var 2>", "Print the input numbers", ButtonGui },
    { "Return", "<var 1> <var 2>", "Print the input numbers", ReturnGui },
    { "ShowNum", "<var 1> <var 2>", "Print the input numbers", ShowNumber },
    {0,0,0,0}
};
```

內部執行的程式名稱

外部呼叫的文字字串名稱

硬體橋接方法



可以使用J9 上面的
Pin30,Pin32跳接

Thank You

AVNET[®]

